

BFS Traversal

Step 1: Start at A (Seeking target node K)

Legend



Complete



Processing



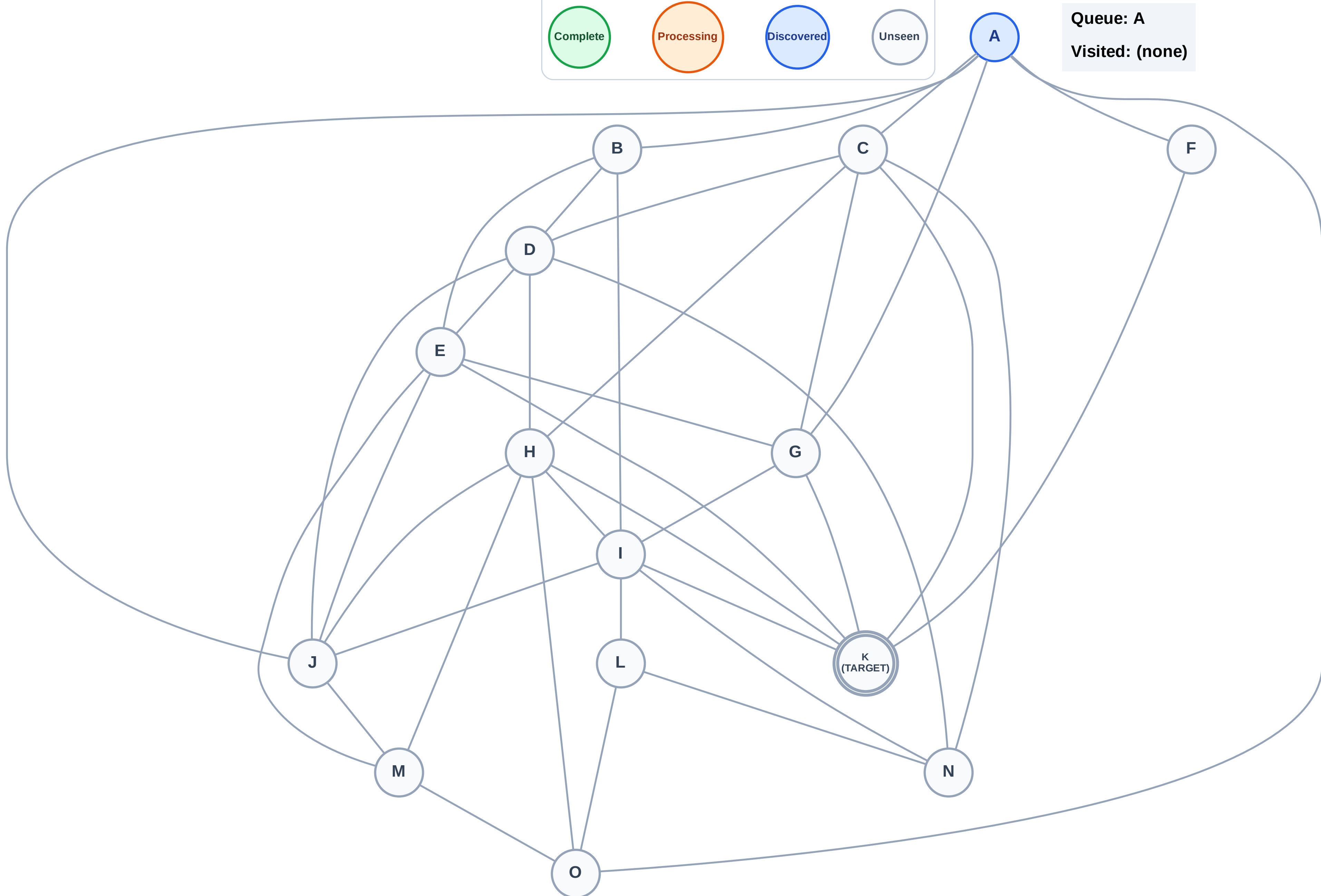
Discovered



Unseen

Queue: A

Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

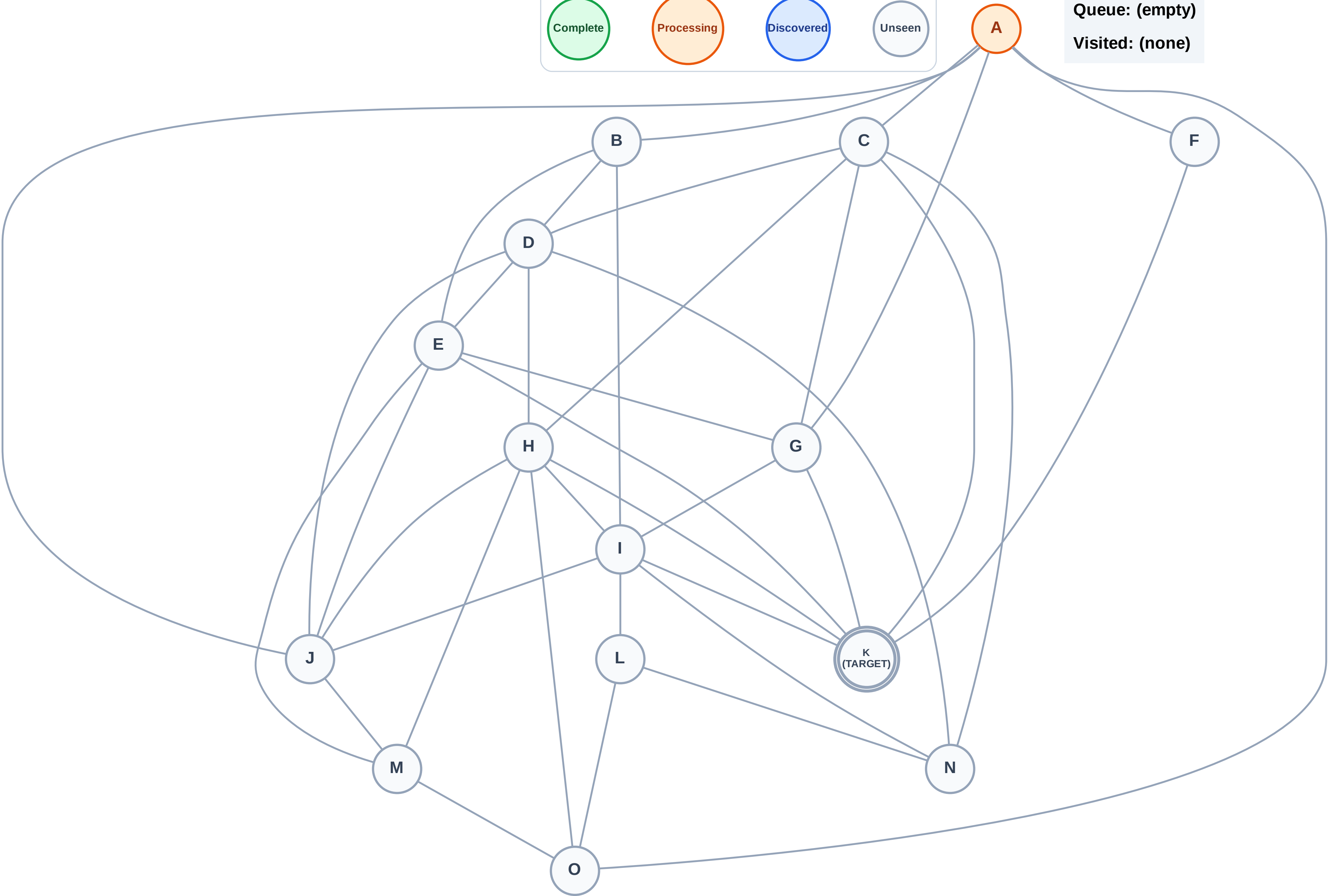
Complete

Processing

Discovered

Unseen

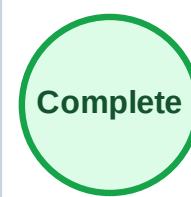
Queue: (empty)
Visited: (none)



BFS Traversal

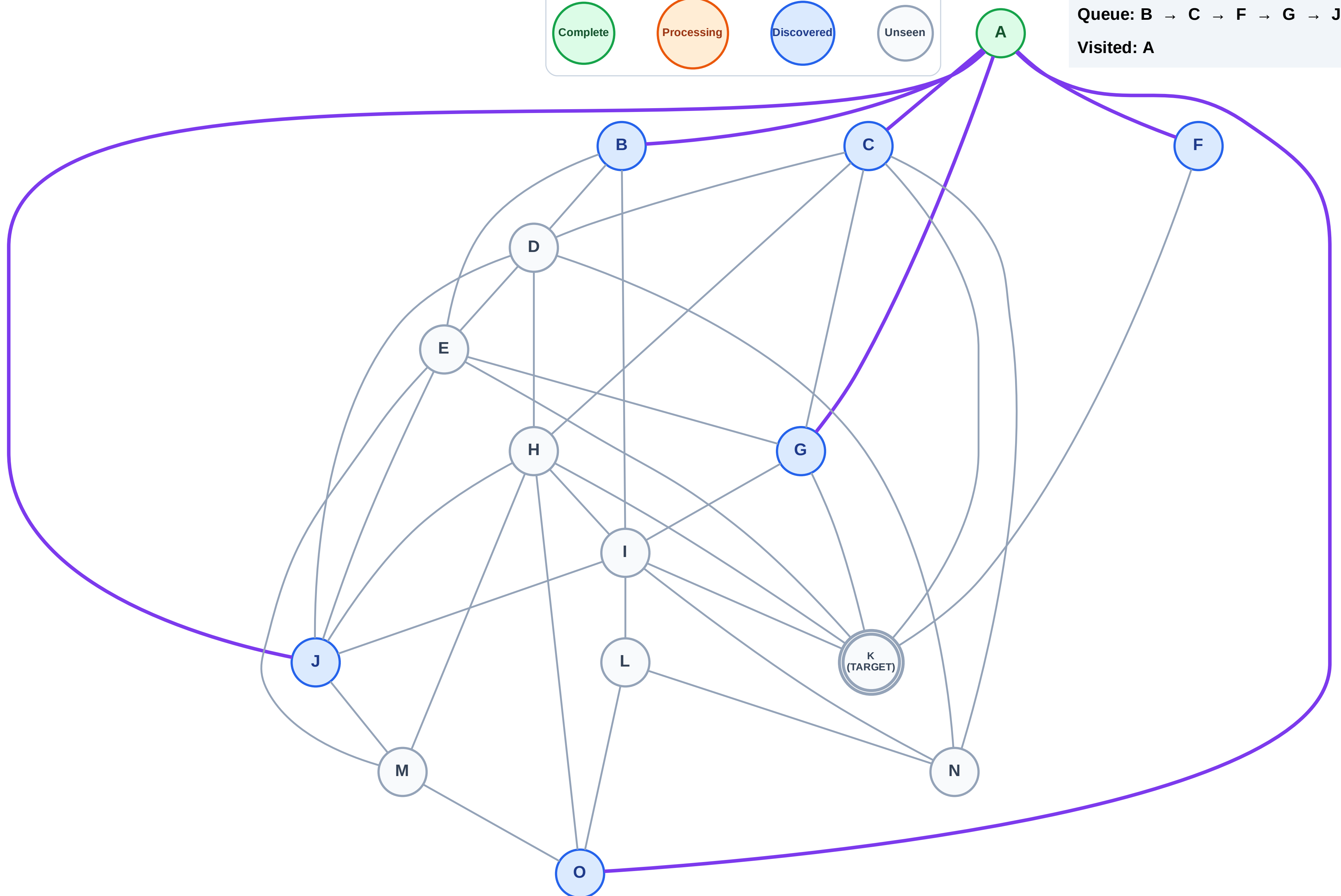
Step 3: Discover B, C, F, G, J, O from A

Legend



Queue: B → C → F → G → J → O

Visited: A



BFS Traversal

Step 4: Process node B

Legend

Complete

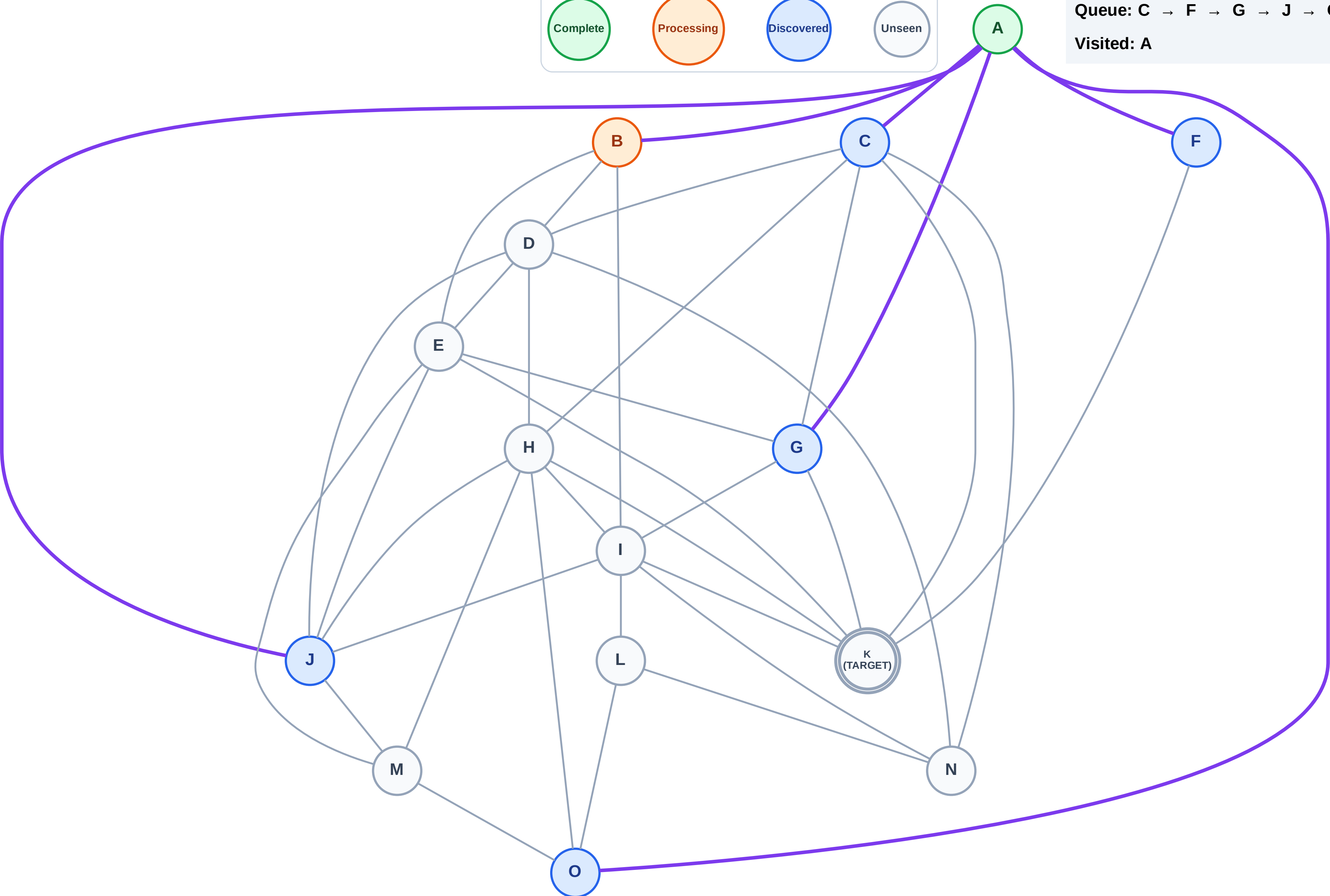
Processing

Discovered

Unseen

Queue: C → F → G → J → O

Visited: A



BFS Traversal

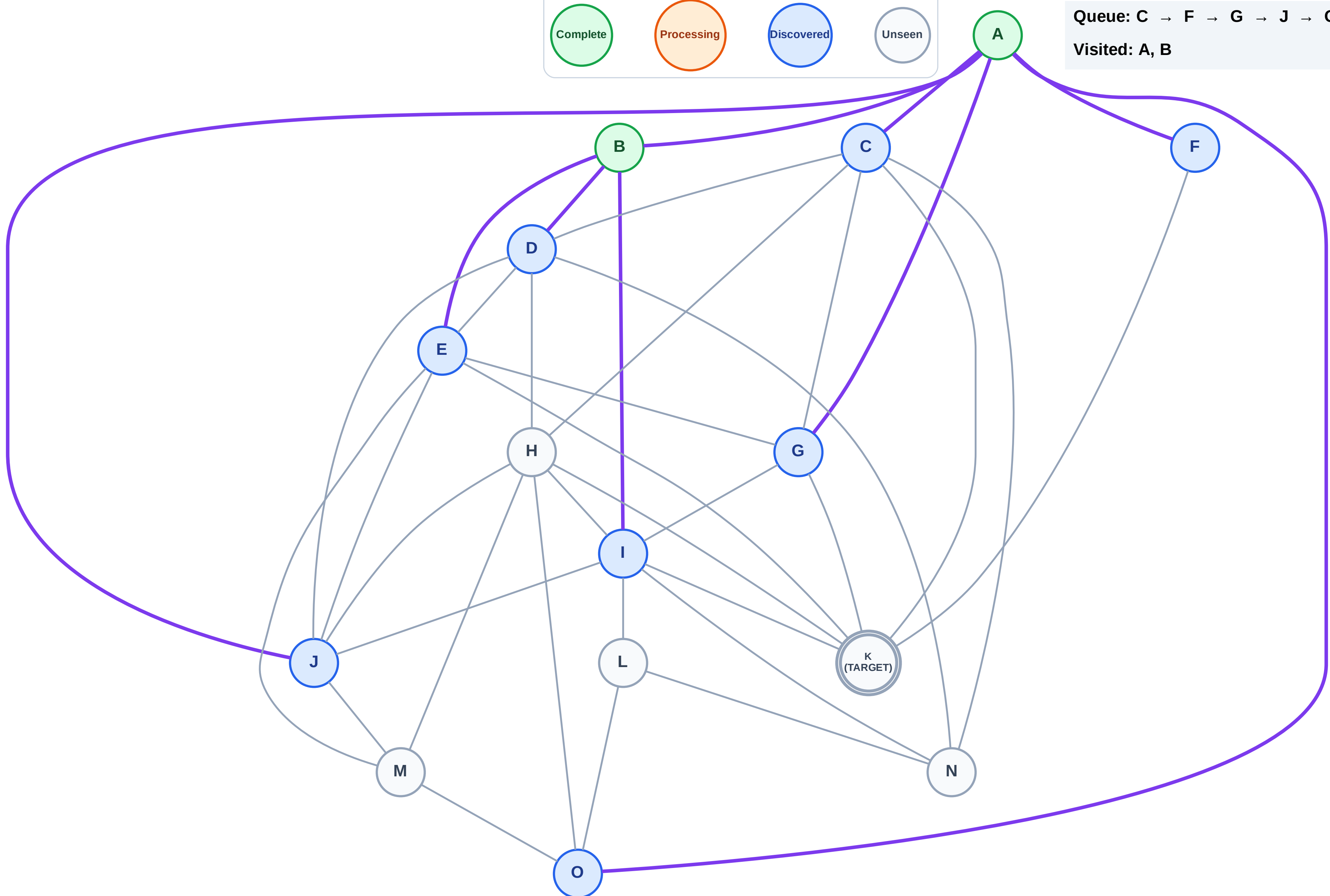
Step 5: Discover D, E, I from B

Legend



Queue: C → F → G → J → O → D → E → I

Visited: A, B



BFS Traversal

Step 6: Process node C

Legend

Complete

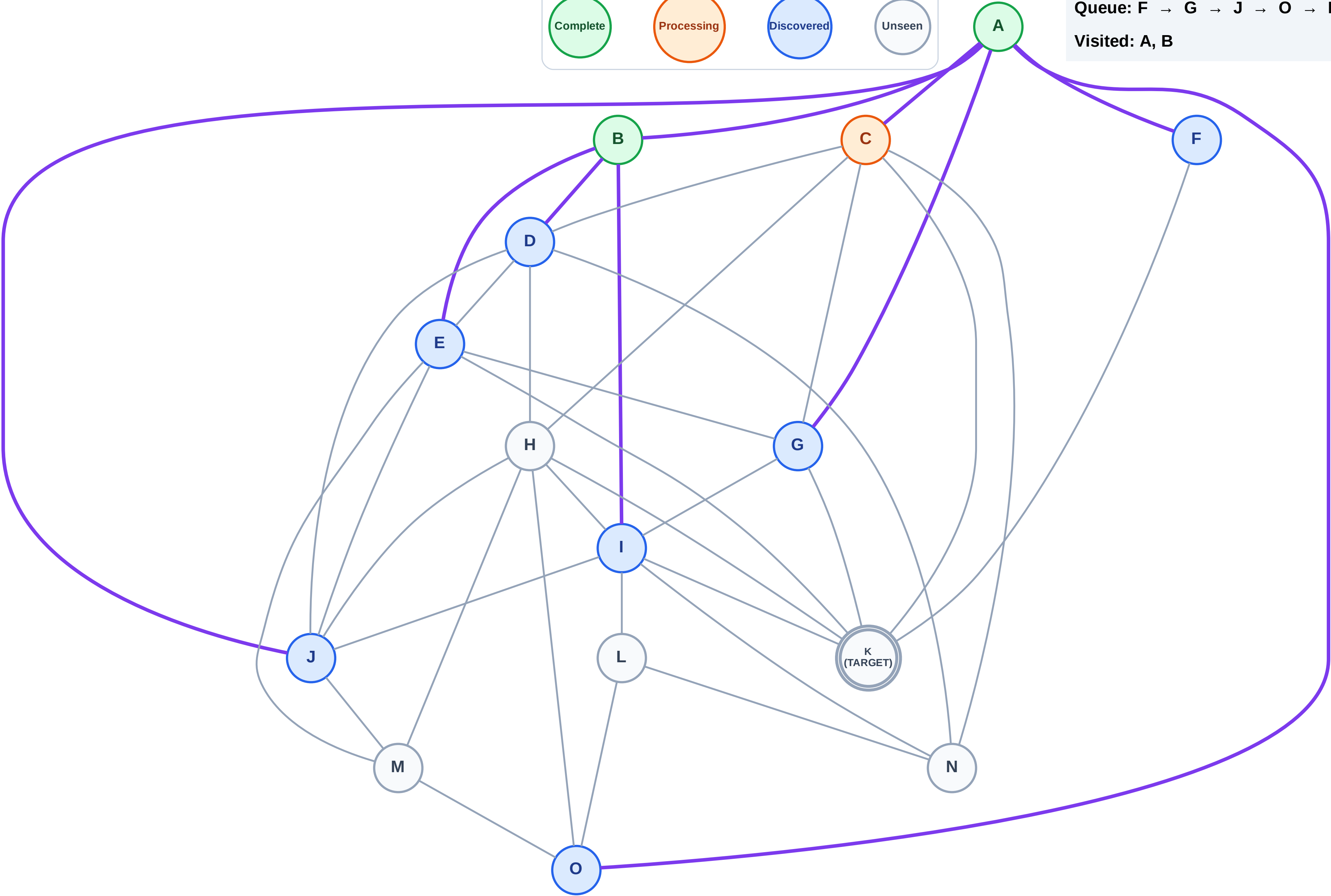
Processing

Discovered

Unseen

Queue: F → G → J → O → D → E → I

Visited: A, B



BFS Traversal

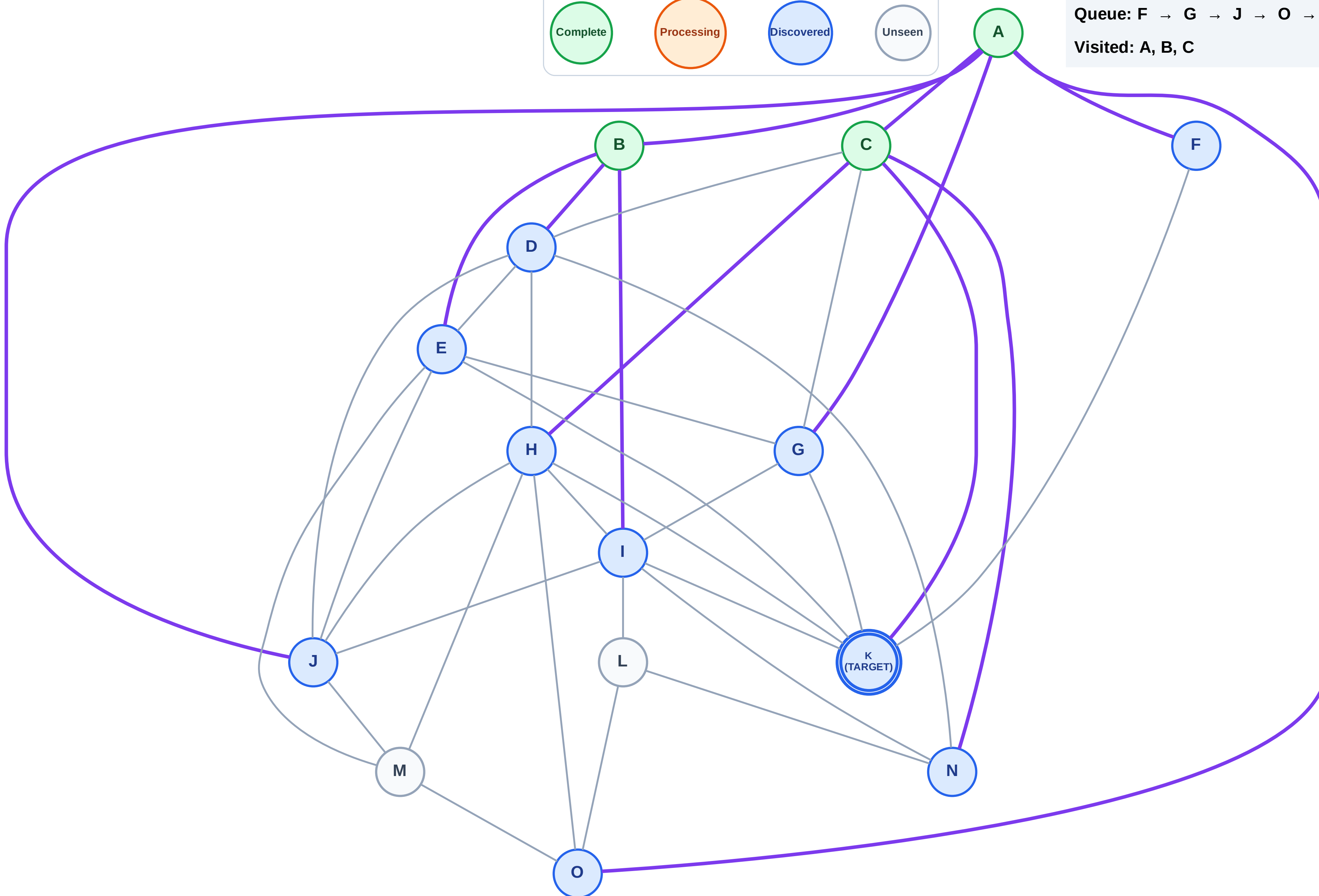
Step 7: Discover H, K, N from C

Legend



Queue: F → G → J → O → D → E → I → H → K → N

Visited: A, B, C



BFS Traversal

Step 8: Process node F

Legend

Complete

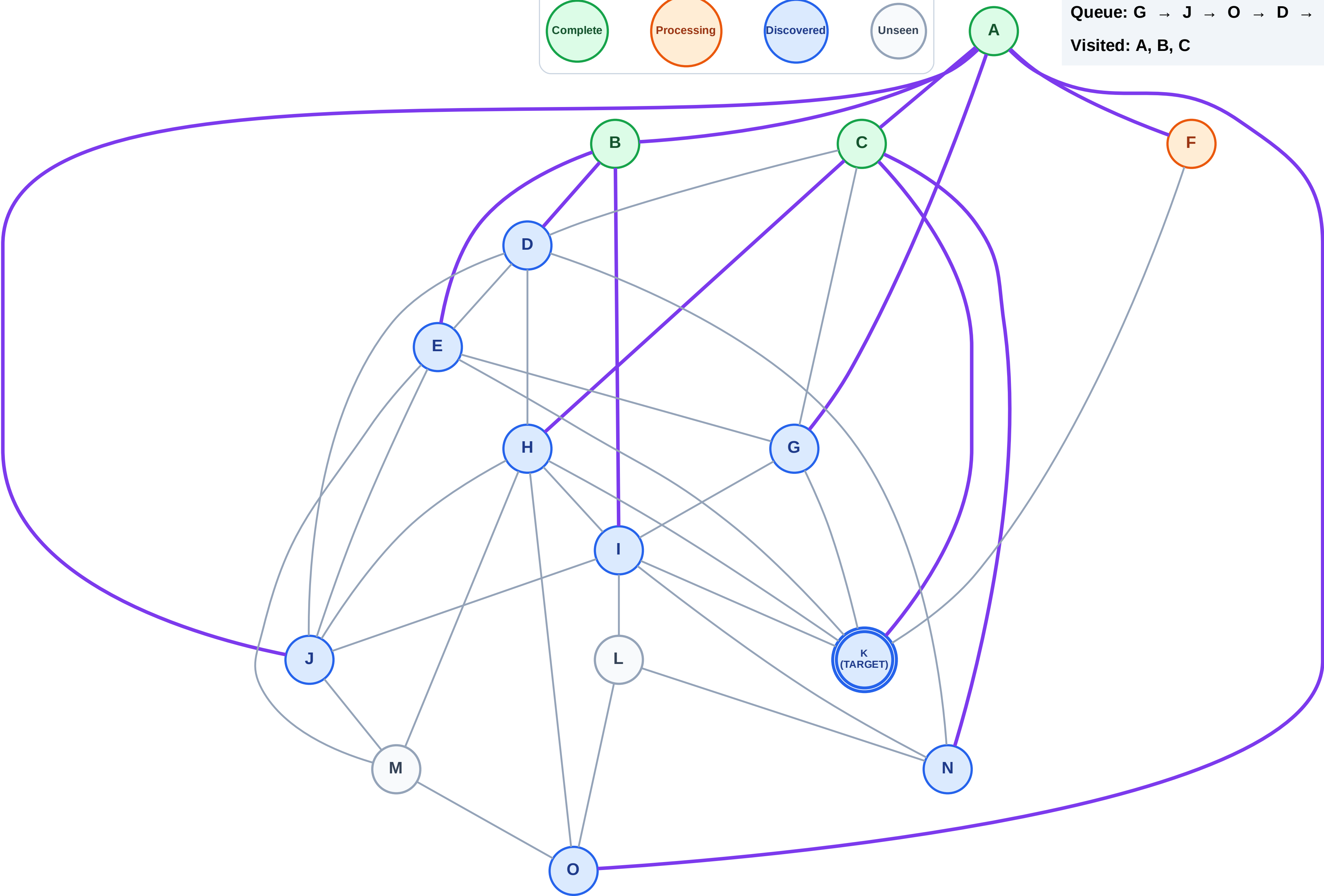
Processing

Discovered

Unseen

Queue: G → J → O → D → E → I → H → K → N

Visited: A, B, C



BFS Traversal

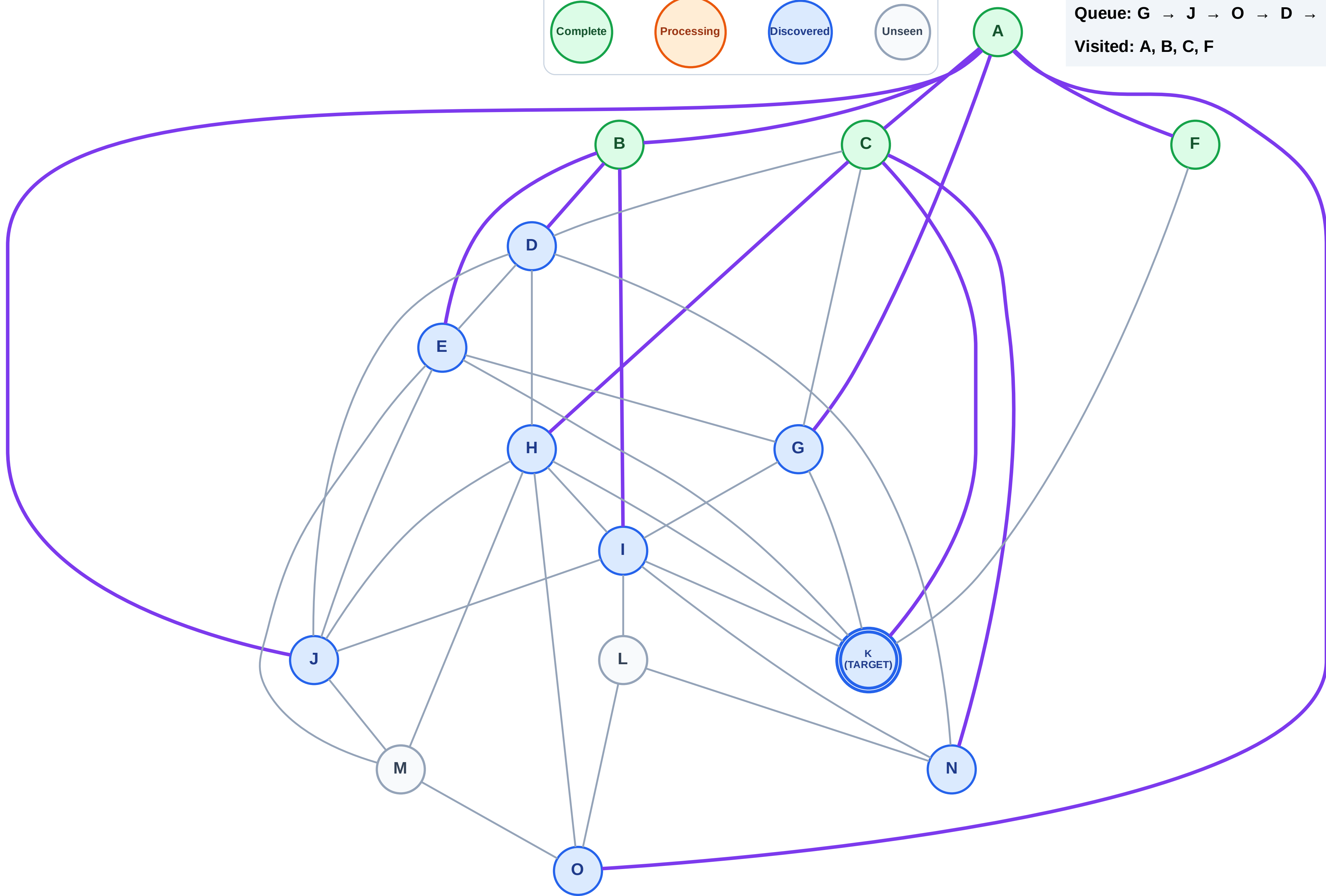
Step 9: No new neighbors from F

Legend



Queue: G → J → O → D → E → I → H → K → N

Visited: A, B, C, F



BFS Traversal

Step 10: Process node G

Legend



Complete



Processing



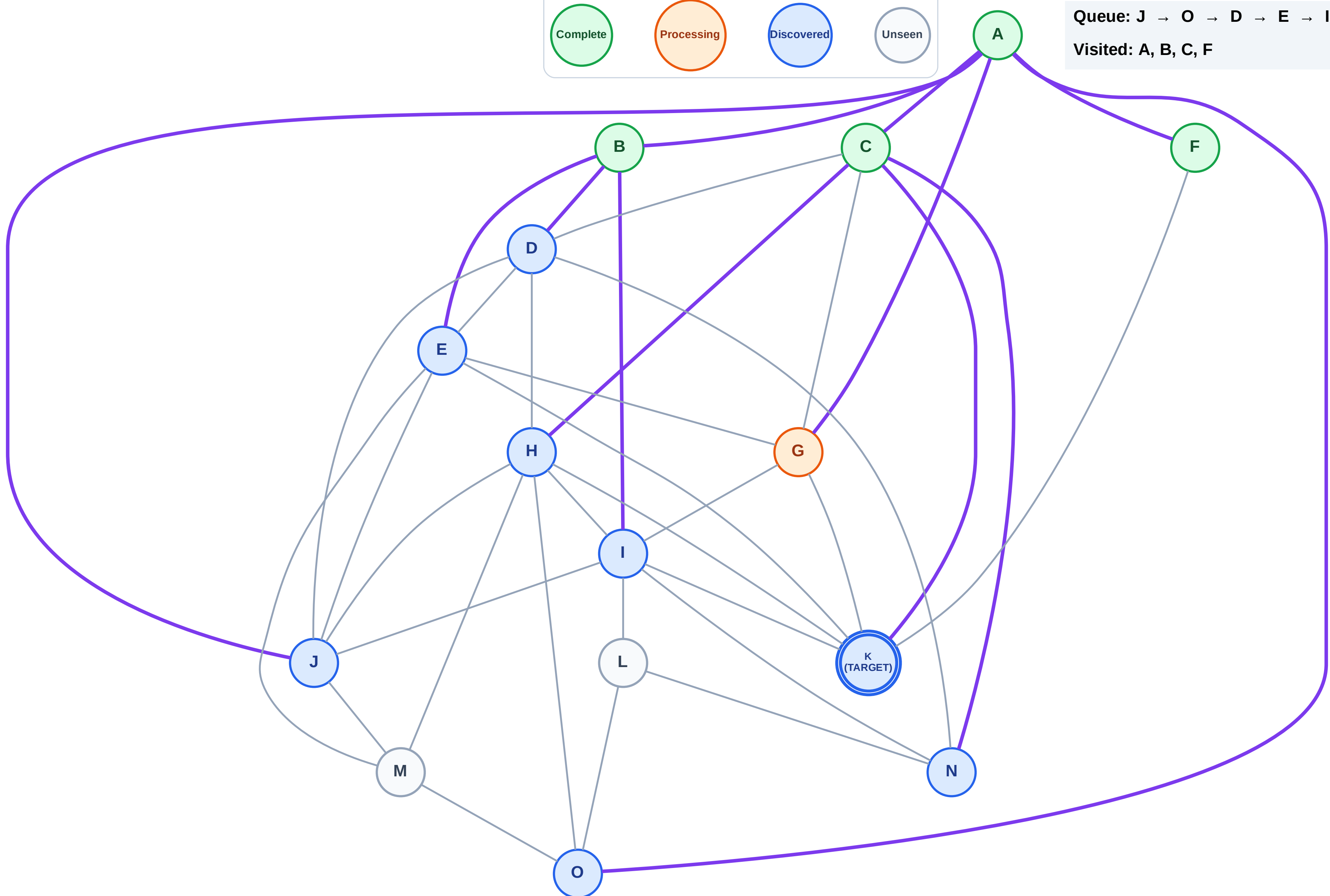
Discovered



Unseen

Queue: J → O → D → E → I → H → K → N

Visited: A, B, C, F



BFS Traversal

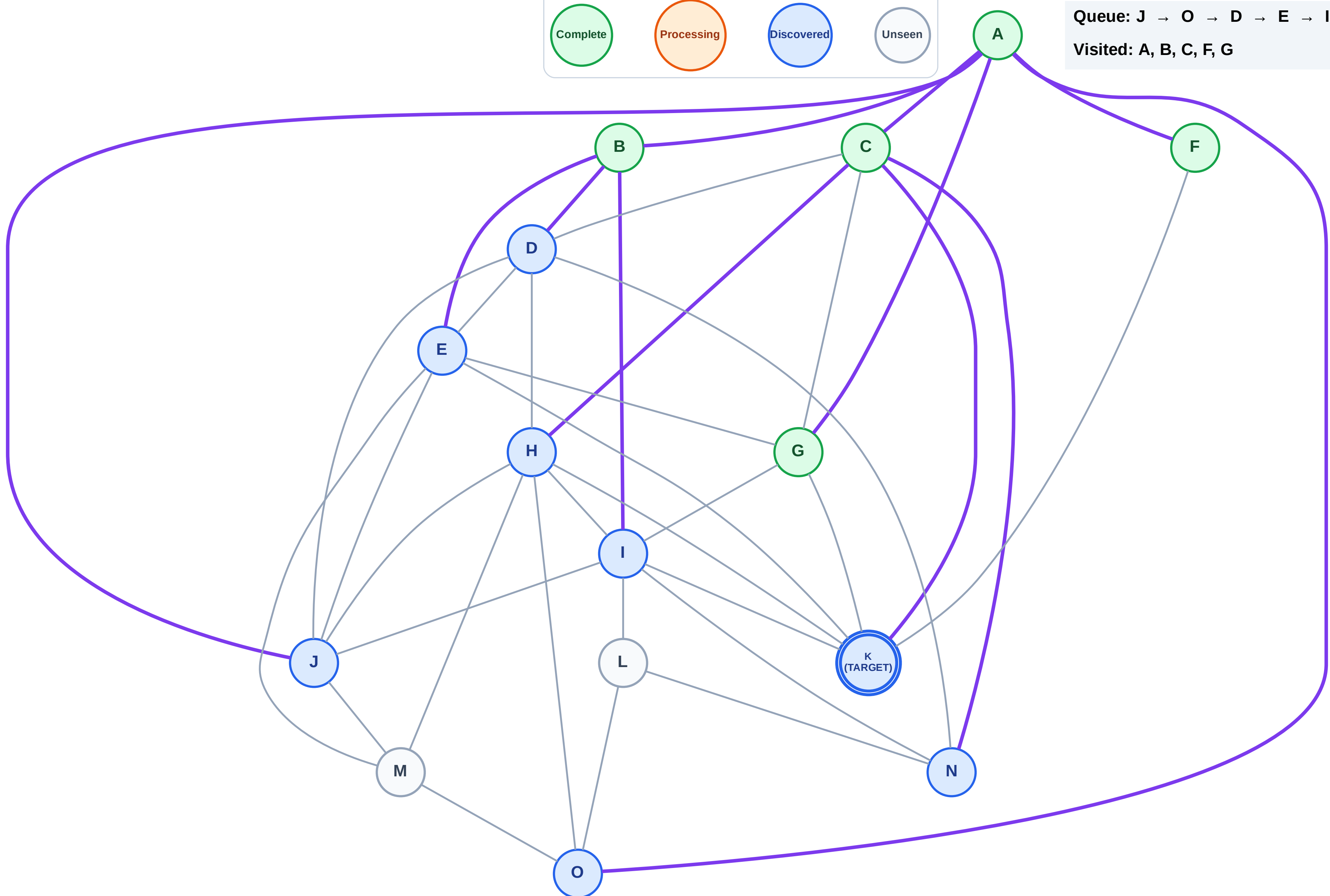
Step 11: No new neighbors from G

Legend



Queue: J → O → D → E → I → H → K → N

Visited: A, B, C, F, G



BFS Traversal

Step 12: Process node J

Legend

Complete

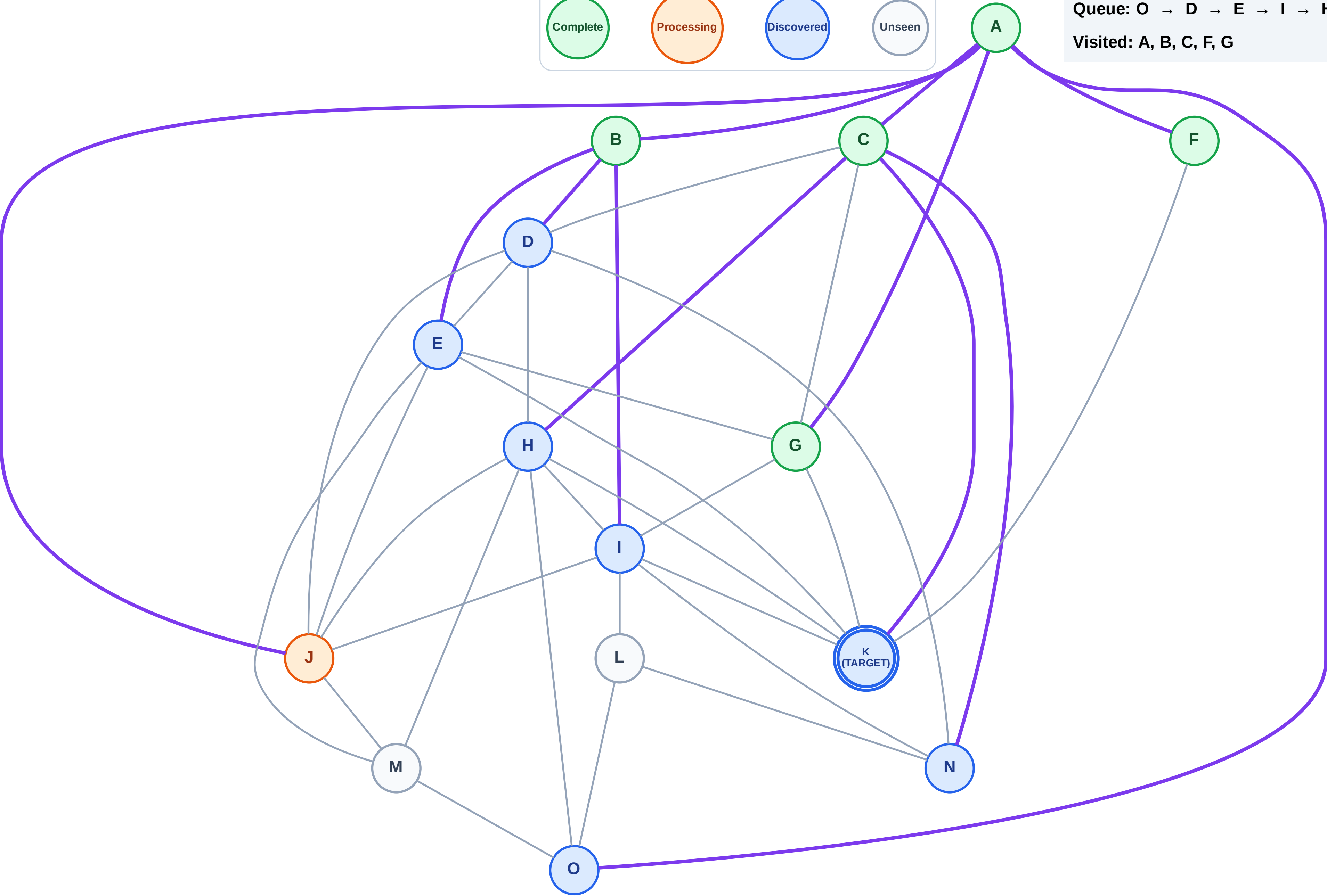
Processing

Discovered

Unseen

Queue: O → D → E → I → H → K → N

Visited: A, B, C, F, G



BFS Traversal

Step 13: Discover M from J

Legend



Complete



Processing



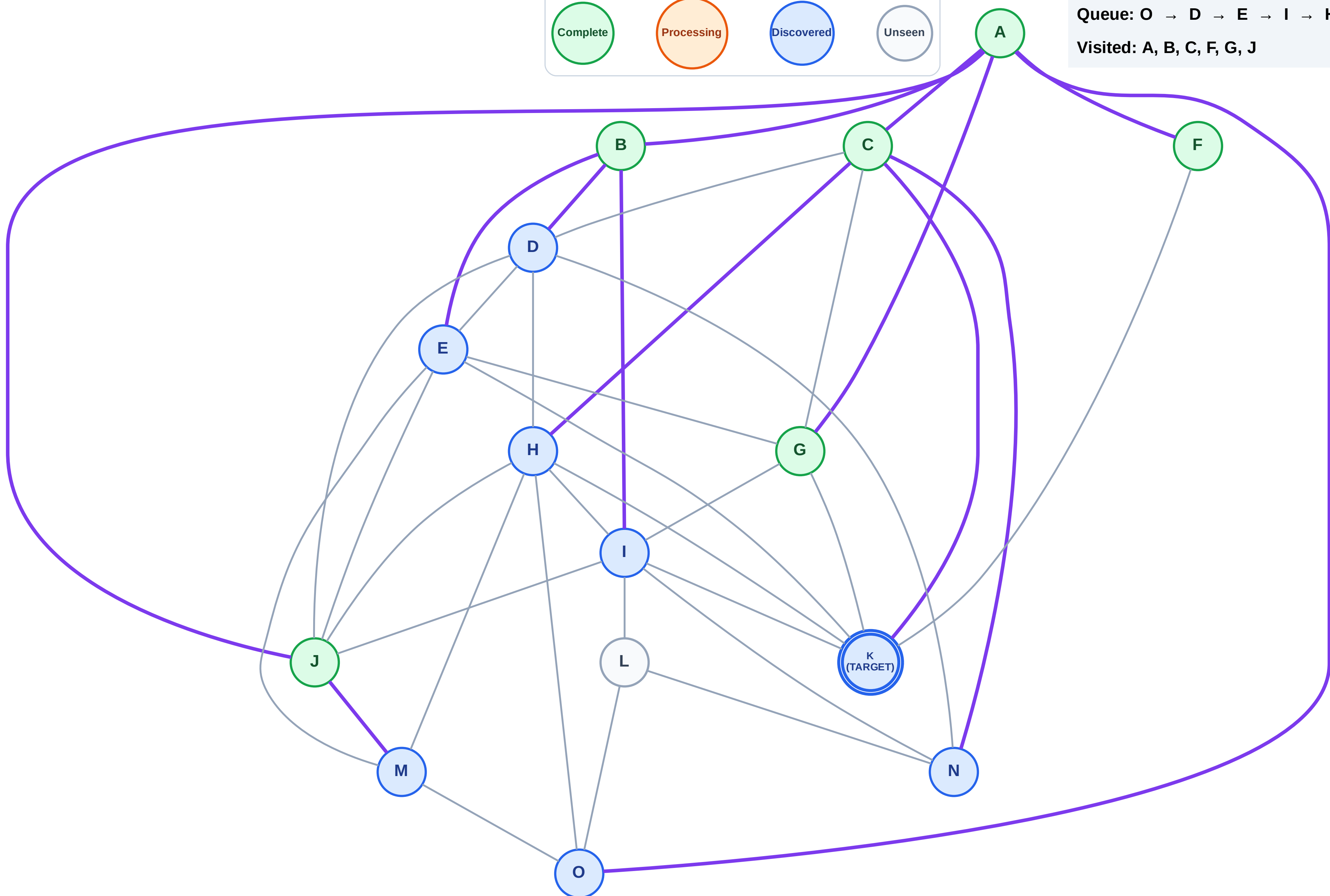
Discovered



Unseen

Queue: O → D → E → I → H → K → N → M

Visited: A, B, C, F, G, J



BFS Traversal

Step 14: Process node O

Legend

Complete

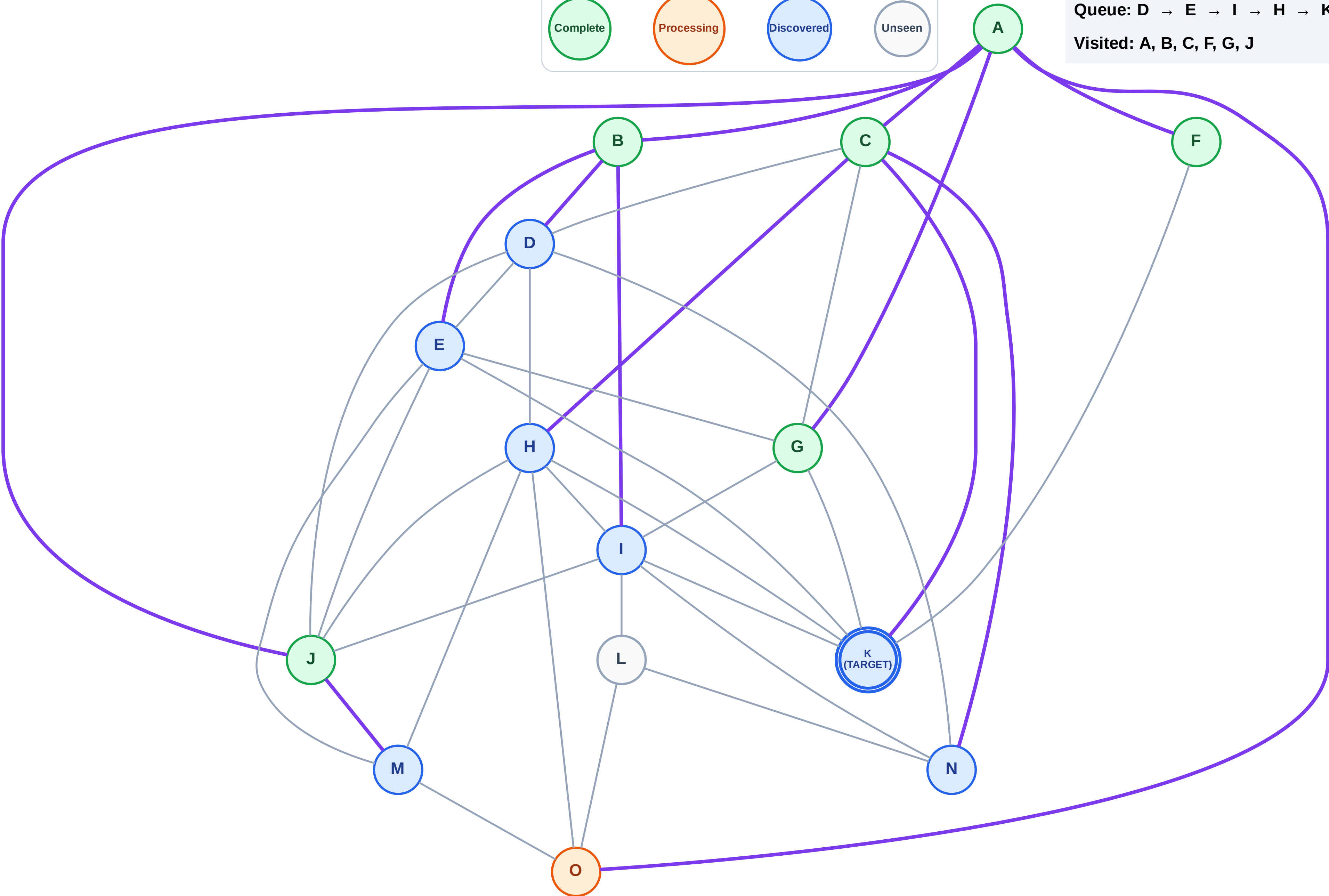
Processing

Discovered

Unseen

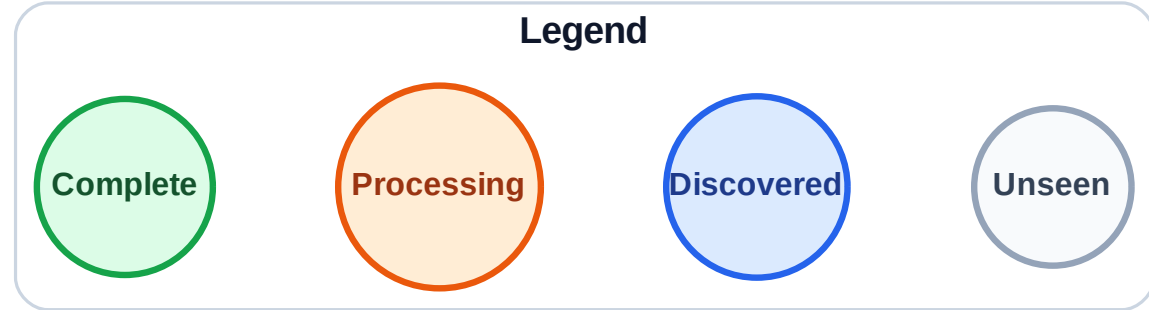
Queue: D → E → I → H → K → N → M

Visited: A, B, C, F, G, J



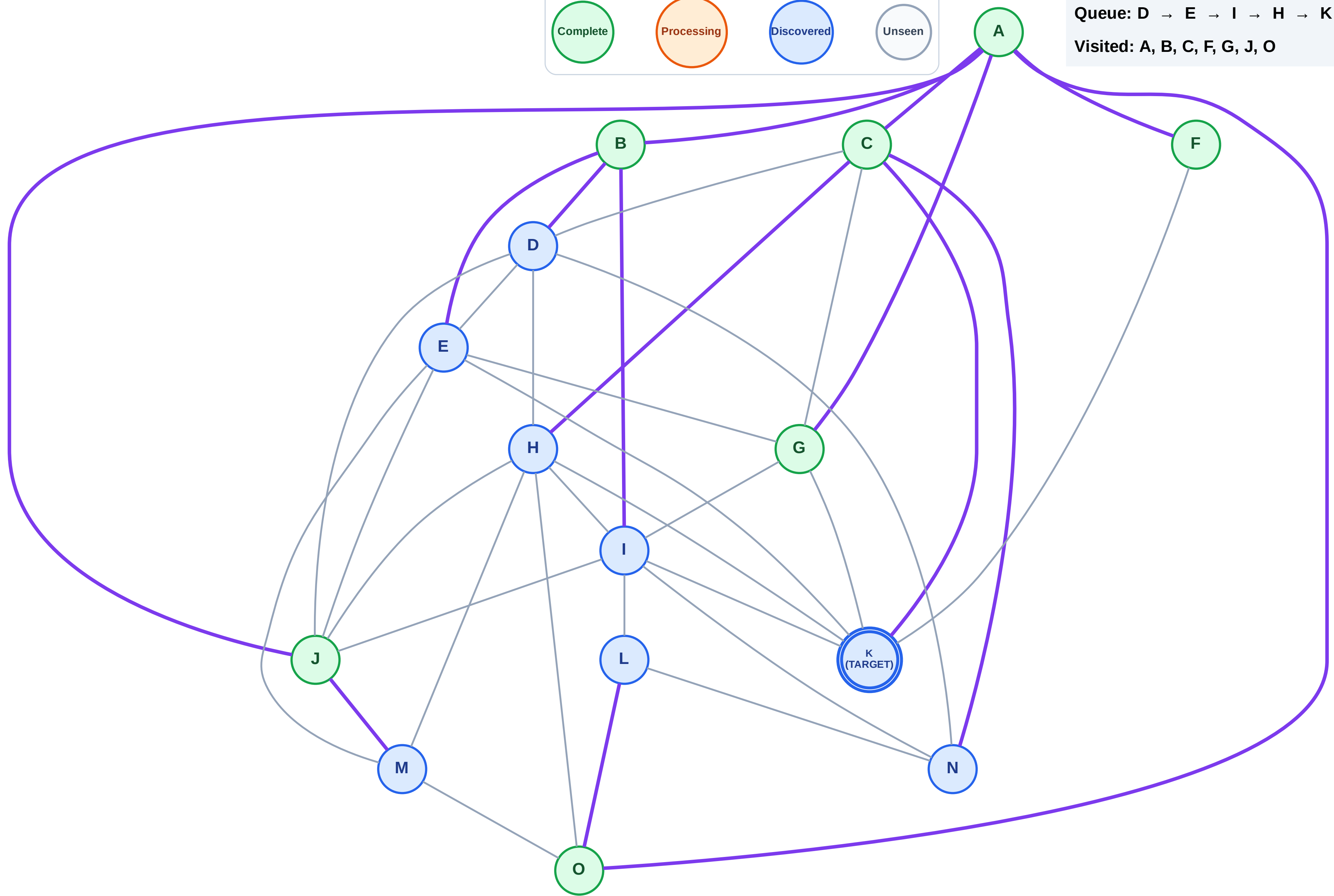
BFS Traversal

Step 15: Discover L from O



Queue: D → E → I → H → K → N → M → L

Visited: A, B, C, F, G, J, O



BFS Traversal

Step 16: Process node D

Legend

Complete

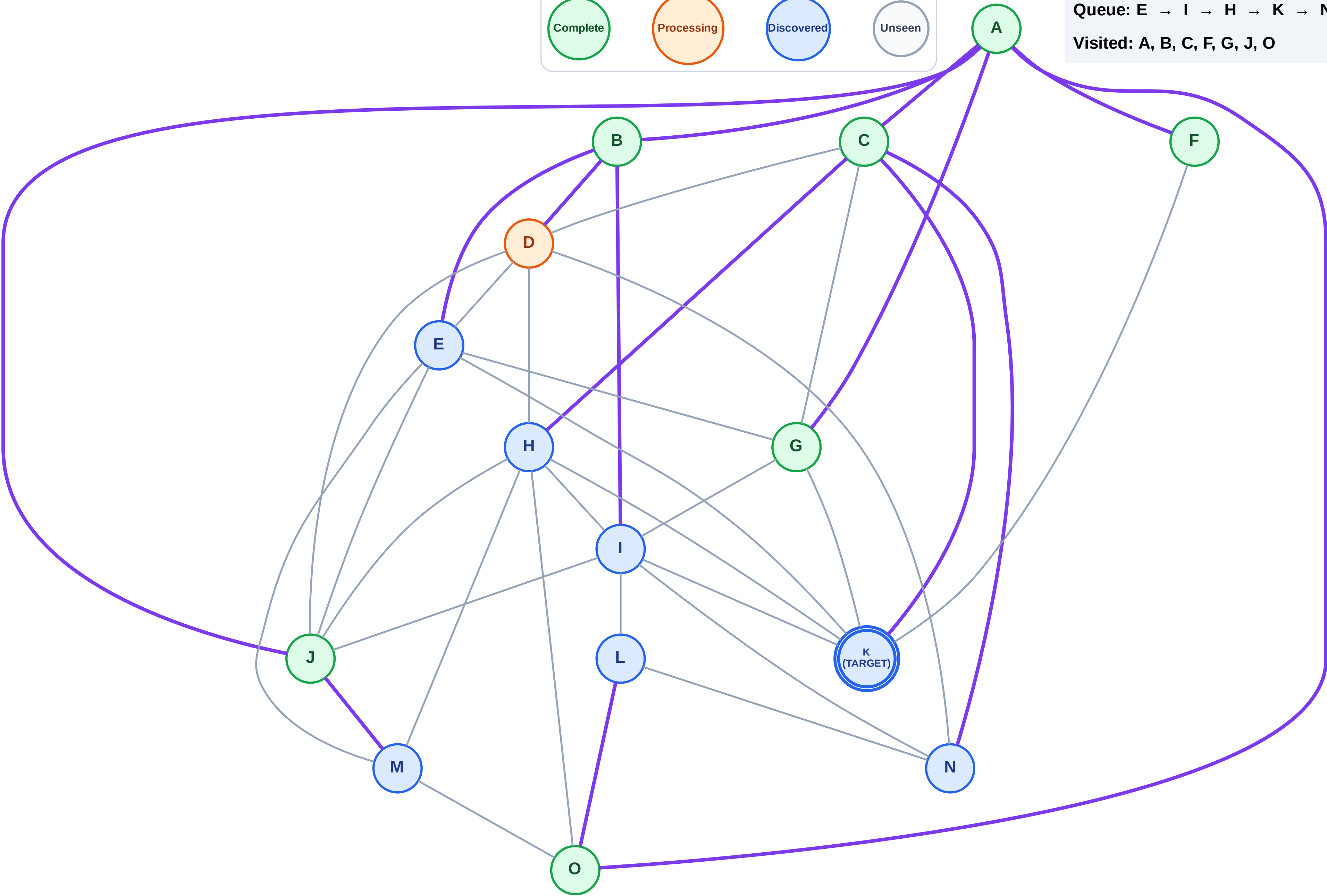
Processing

Discovered

Unseen

Queue: E → I → H → K → N → M → L

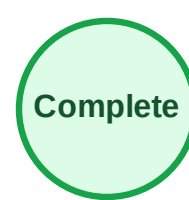
Visited: A, B, C, F, G, J, O



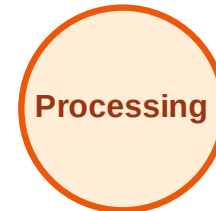
BFS Traversal

Step 17: No new neighbors from D

Legend



Complete



Processing



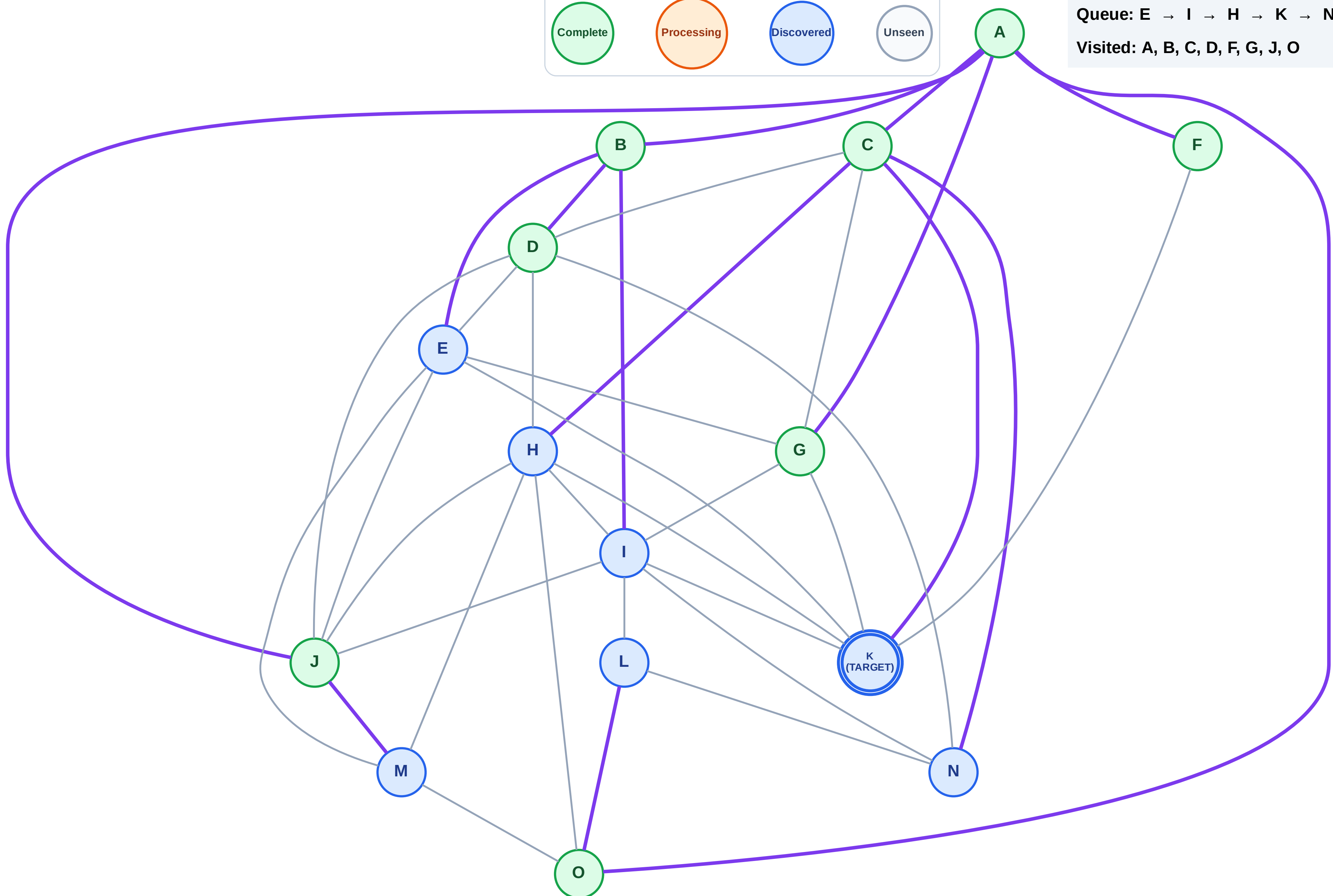
Discovered



Unseen

Queue: E → I → H → K → N → M → L

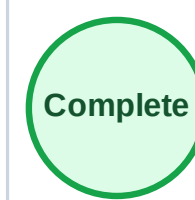
Visited: A, B, C, D, F, G, J, O



BFS Traversal

Step 18: Process node E

Legend



Complete



Processing



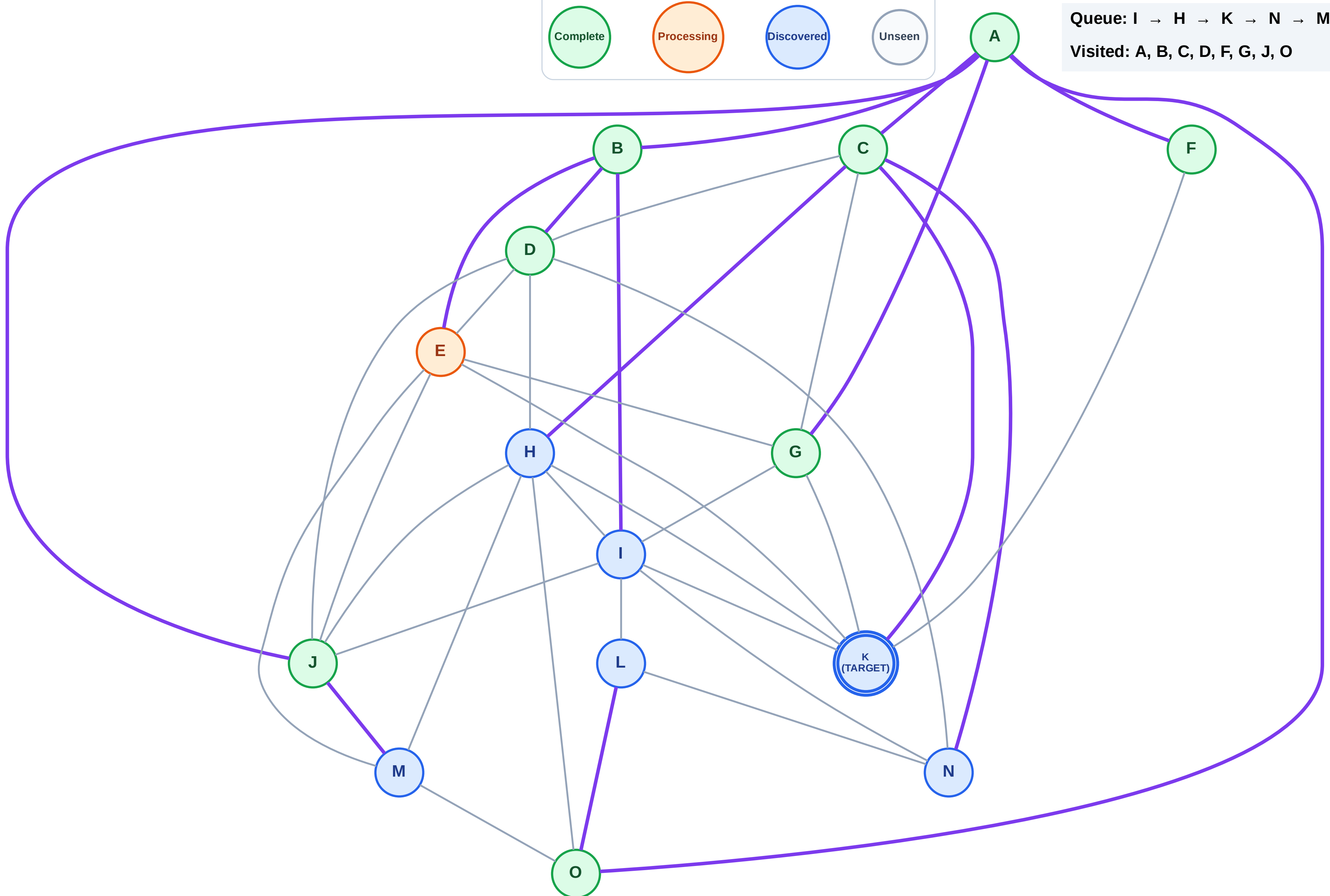
Discovered



Unseen

Queue: I → H → K → N → M → L

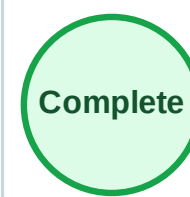
Visited: A, B, C, D, F, G, J, O



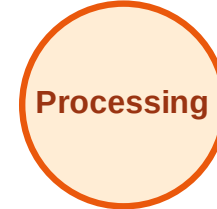
BFS Traversal

Step 19: No new neighbors from E

Legend



Complete



Processing



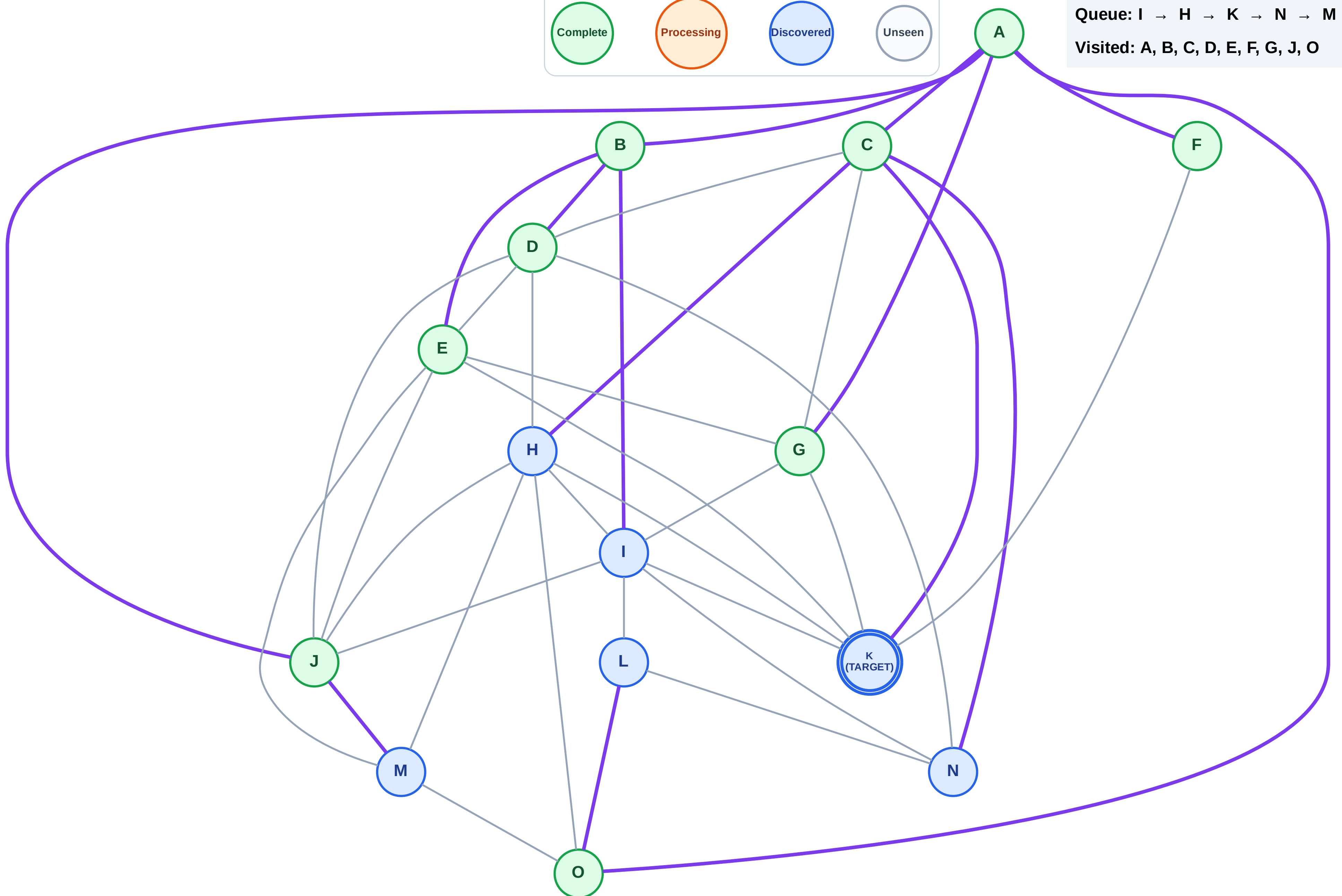
Discovered



Unseen

Queue: I → H → K → N → M → L

Visited: A, B, C, D, E, F, G, J, O



BFS Traversal

Step 20: Process node I

Legend

Complete

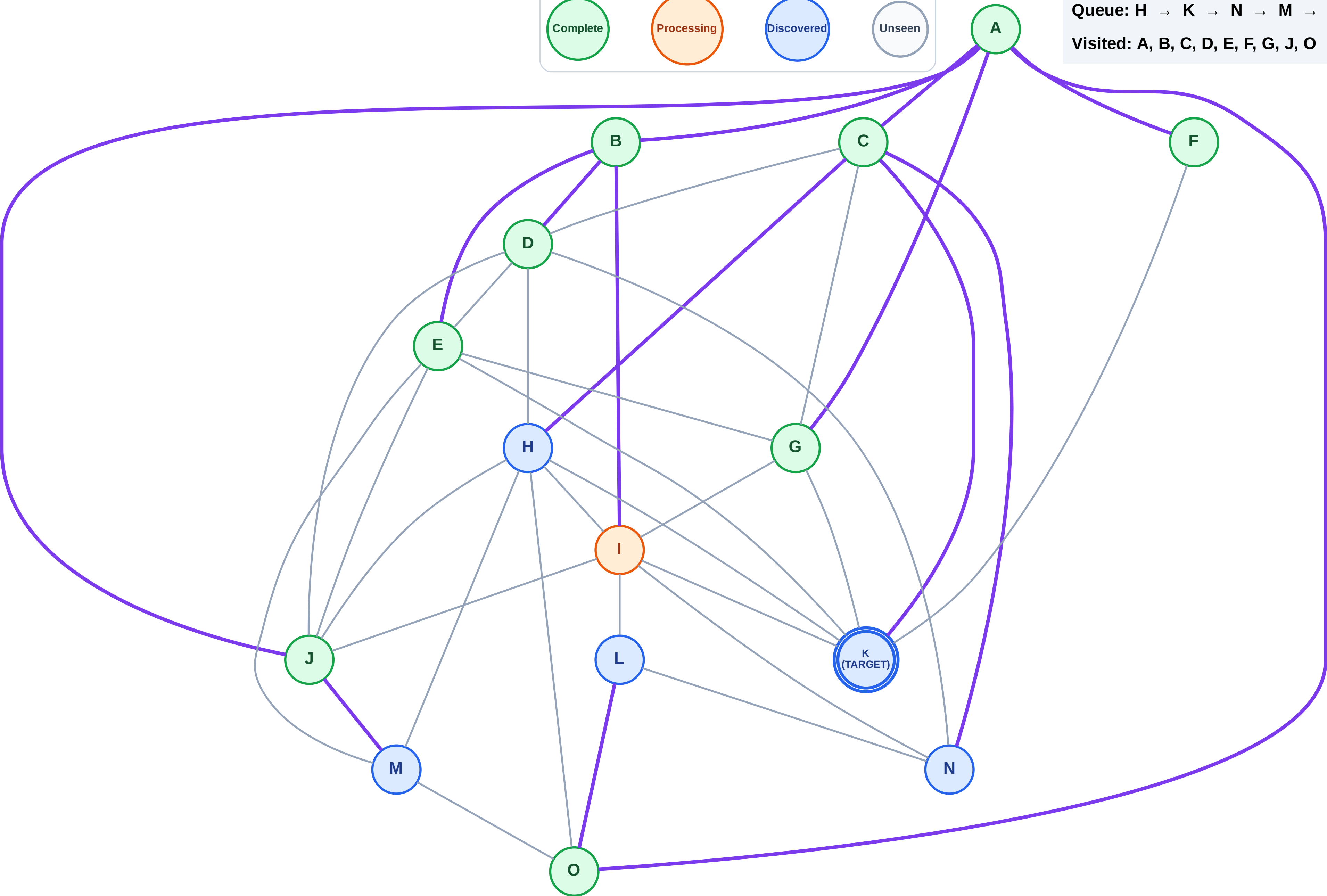
Processing

Discovered

Unseen

Queue: H → K → N → M → L

Visited: A, B, C, D, E, F, G, J, O



BFS Traversal

Step 21: No new neighbors from I

Legend

Complete

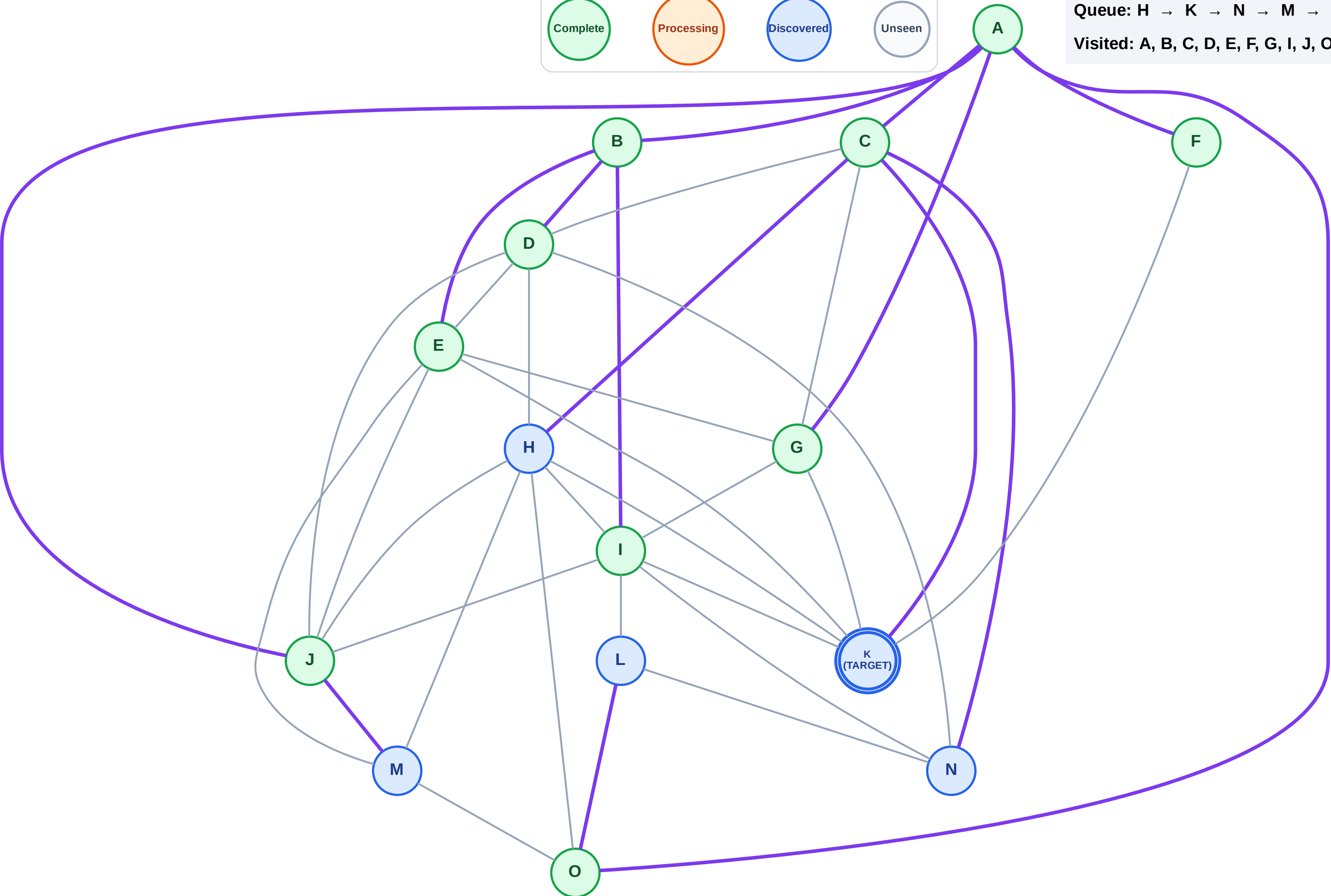
Processing

Discovered

Unseen

Queue: H → K → N → M → L

Visited: A, B, C, D, E, F, G, I, J, O



BFS Traversal

Step 22: Process node H

Legend

Complete

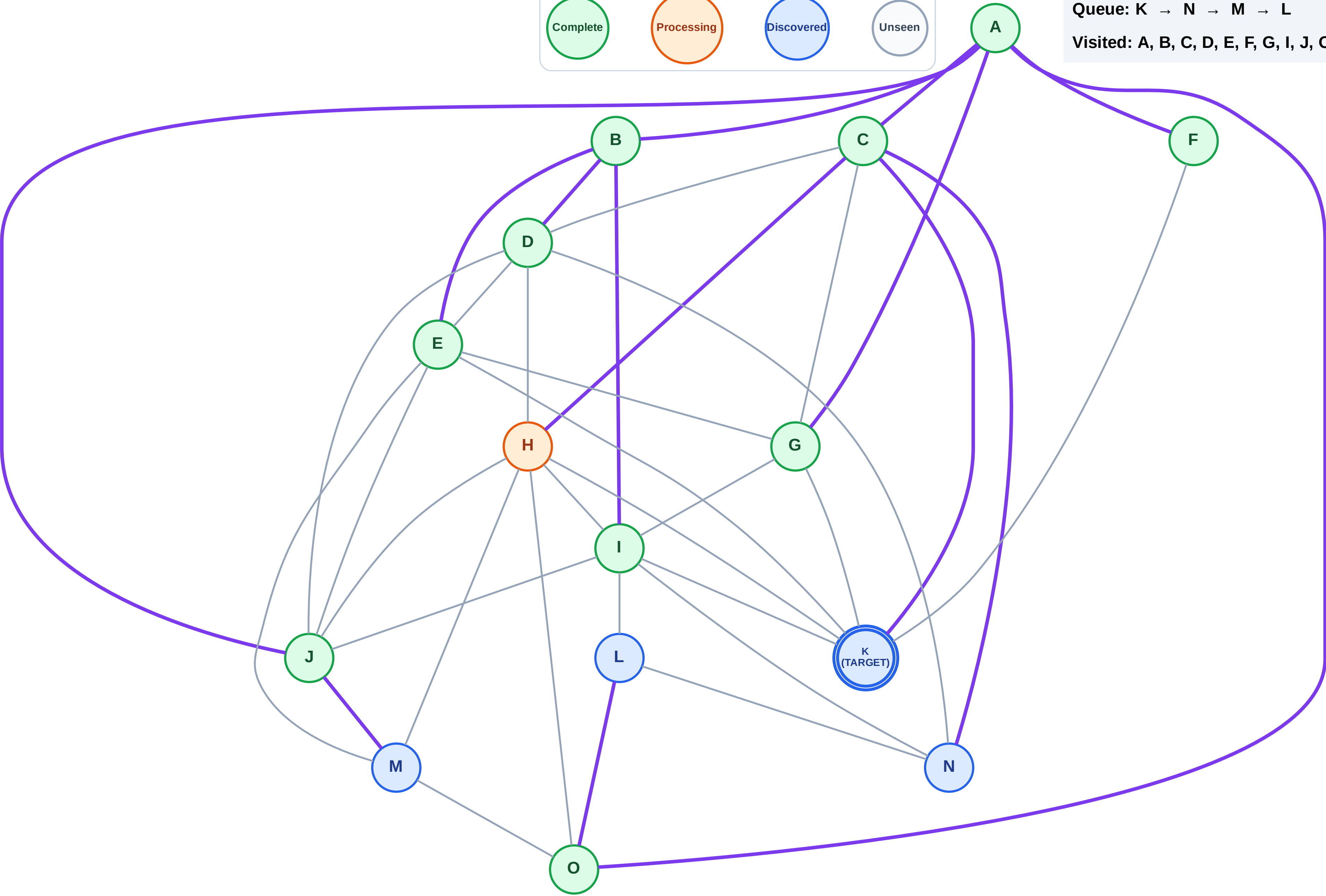
Processing

Discovered

Unseen

Queue: K → N → M → L

Visited: A, B, C, D, E, F, G, I, J, O



BFS Traversal

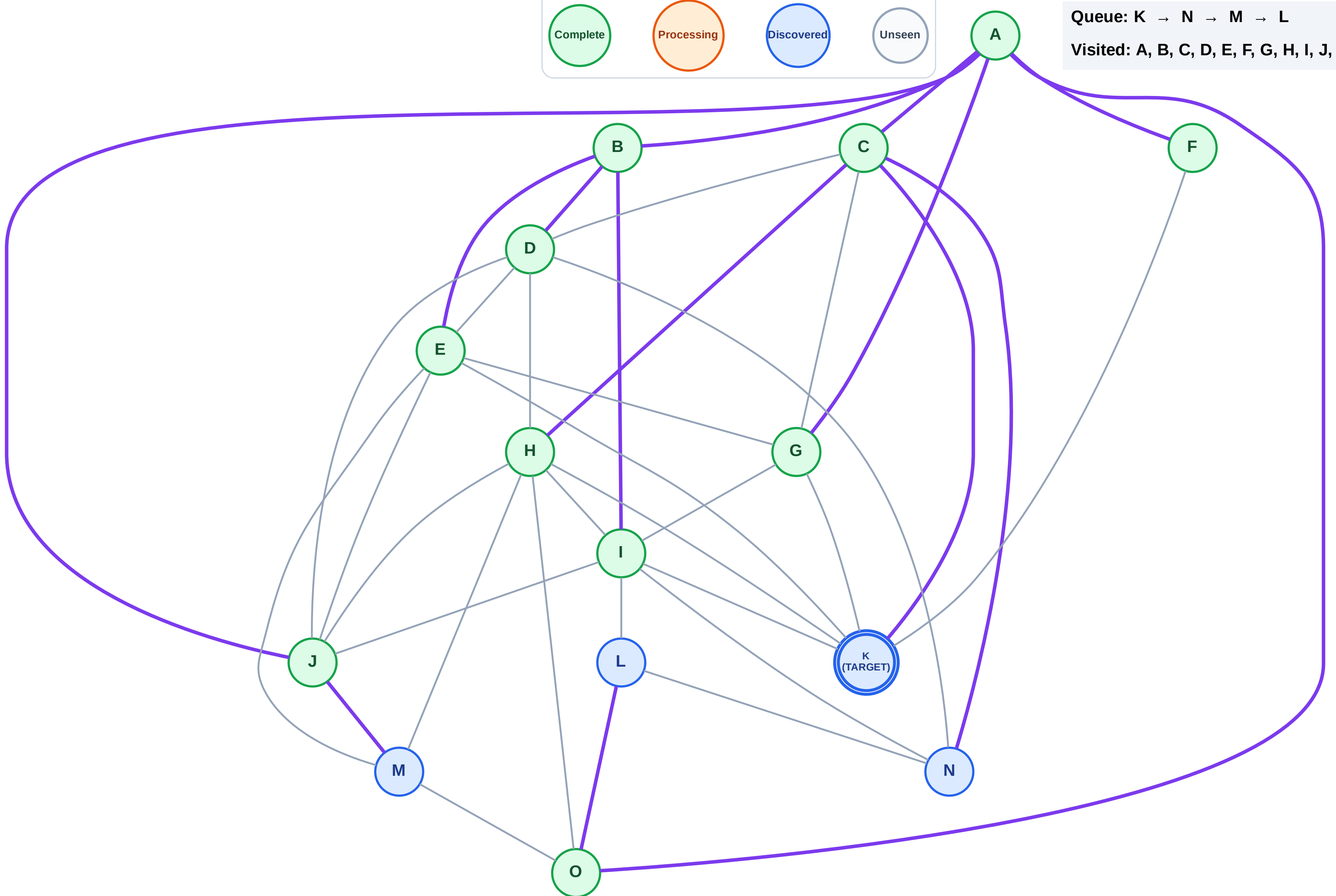
Step 23: No new neighbors from H

Legend



Queue: K → N → M → L

Visited: A, B, C, D, E, F, G, H, I, J, O



BFS Traversal

Step 24: Process node K

Legend

Complete

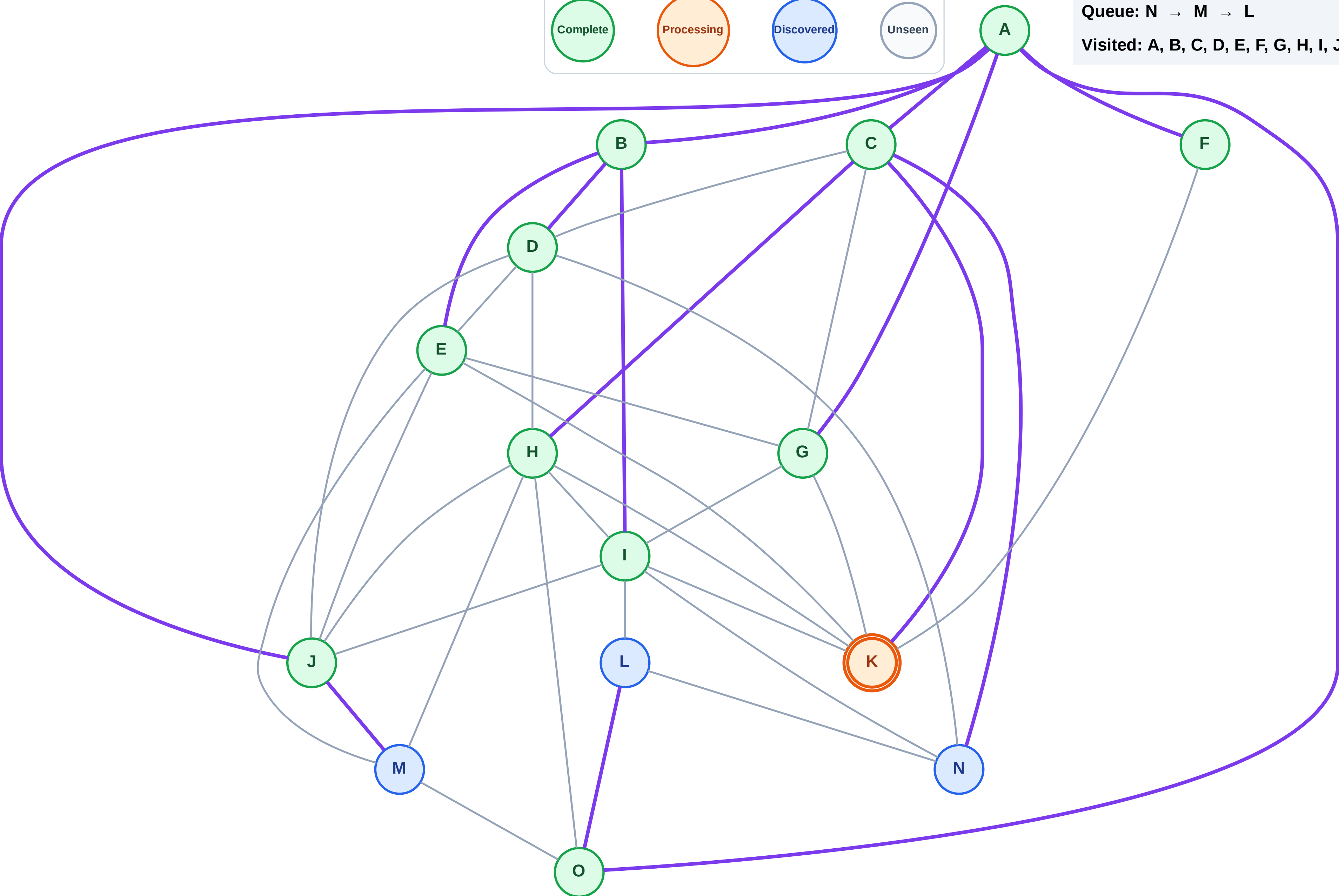
Processing

Discovered

Unseen

Queue: N → M → L

Visited: A, B, C, D, E, F, G, H, I, J, O



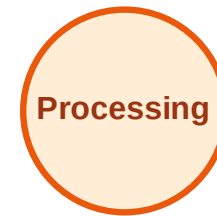
BFS Traversal

Step 25: TARGET NODE K REACHED! Path (2 hops): A → C → K

Legend



Complete



Processing



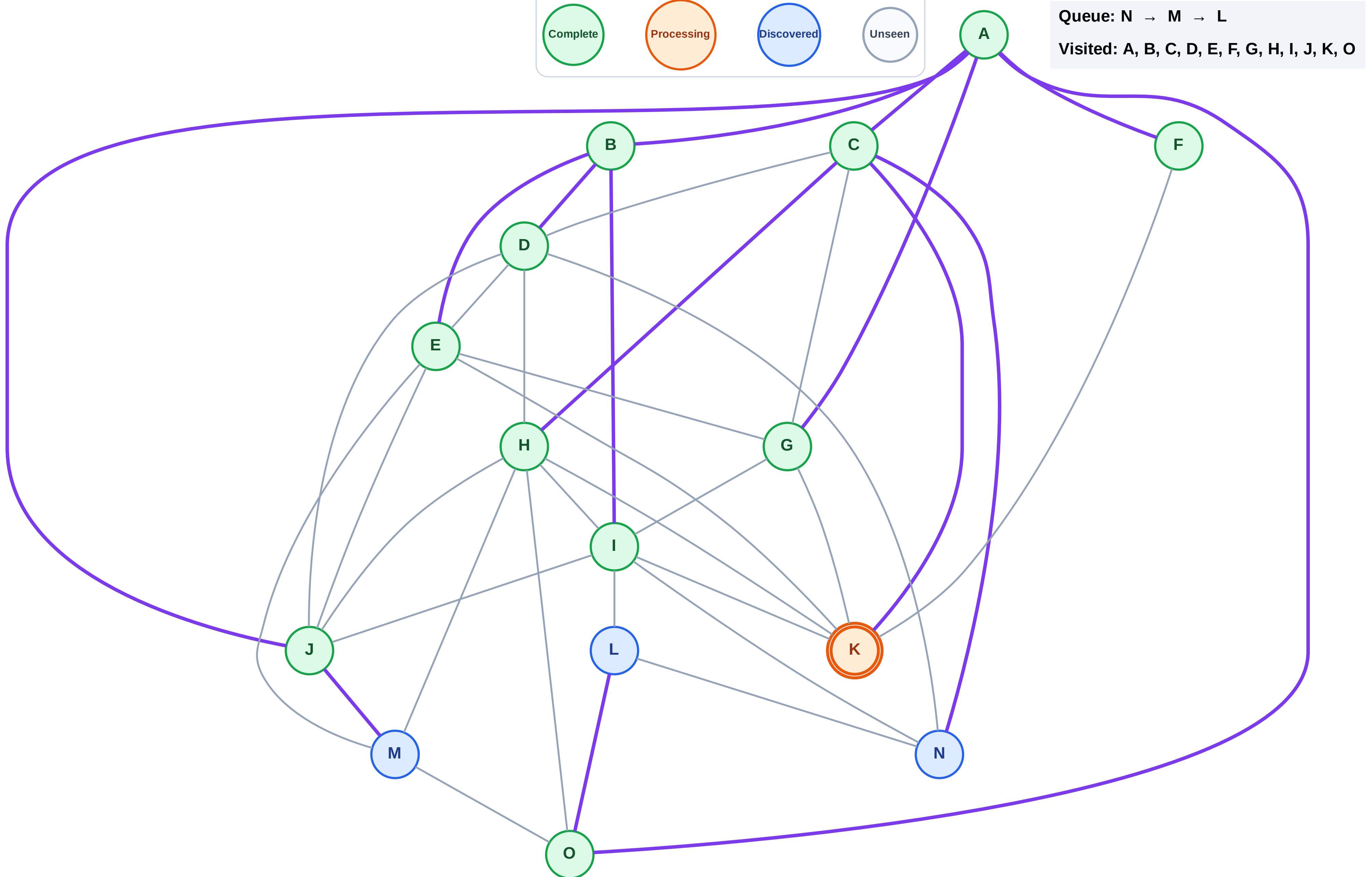
Discovered



Unseen

Queue: N → M → L

Visited: A, B, C, D, E, F, G, H, I, J, K, O



DFS Traversal

Step 1: Start at A (Seeking target node K)

Legend

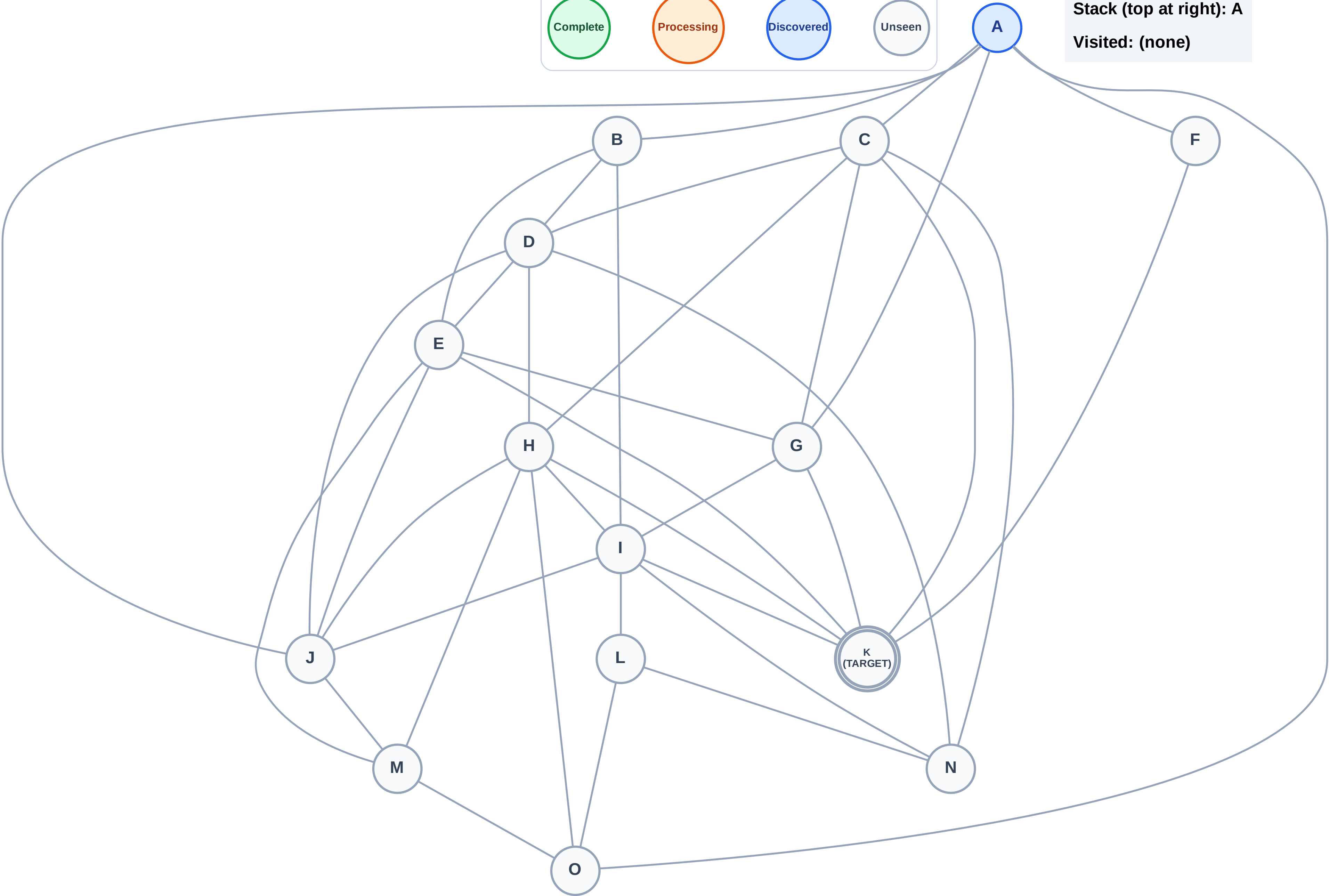
Complete

Processing

Discovered

Unseen

Stack (top at right): A
Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

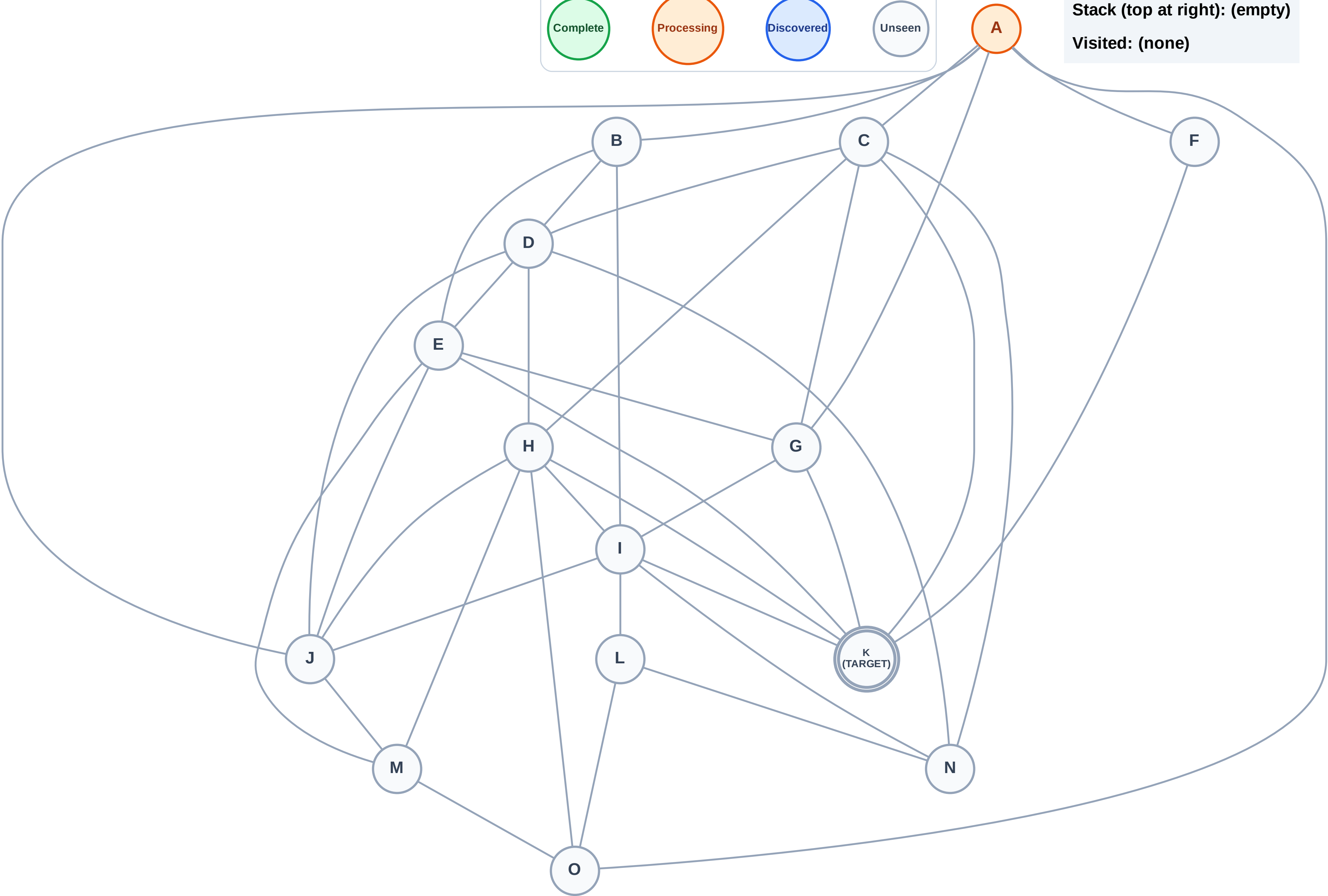
Complete

Processing

Discovered

Unseen

Stack (top at right): (empty)
Visited: (none)



DFS Traversal

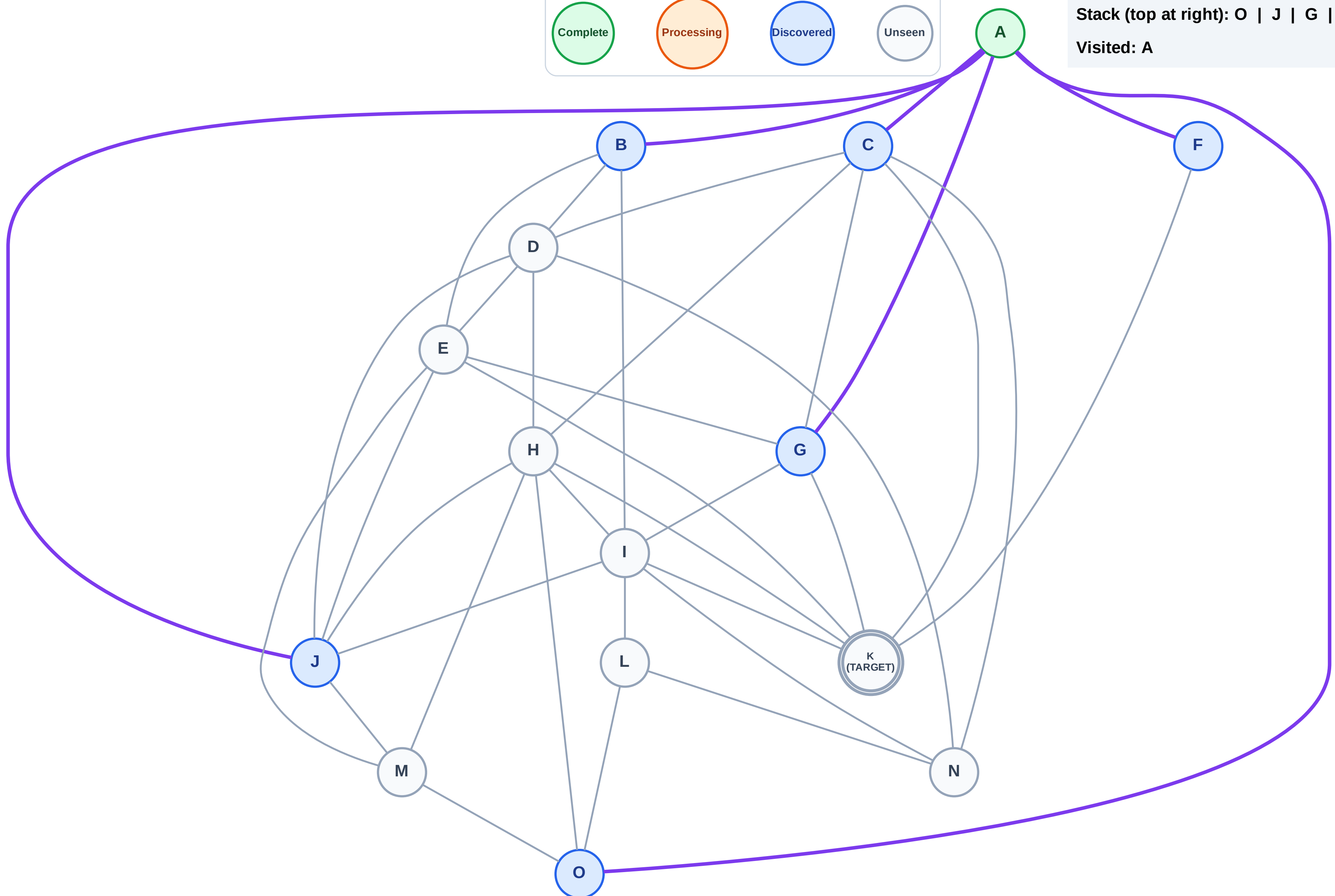
Step 3: Discover O, J, G, F, C, B from A

Legend



Stack (top at right): O | J | G | F | C | B

Visited: A



DFS Traversal

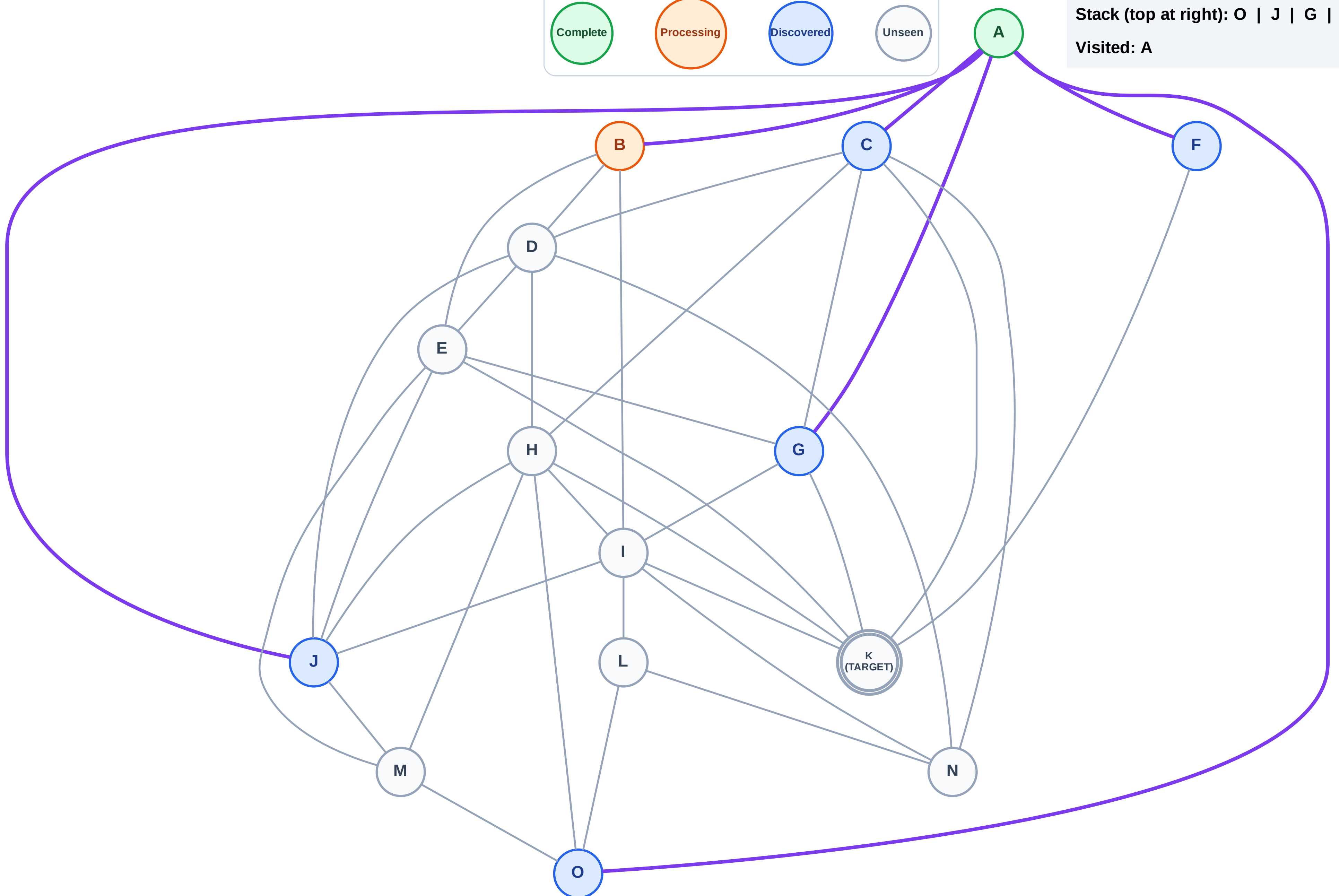
Step 4: Process node B

Legend



Stack (top at right): O | J | G | F | C

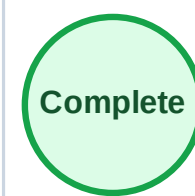
Visited: A



DFS Traversal

Step 5: Discover I, E, D from B

Legend



Complete



Processing



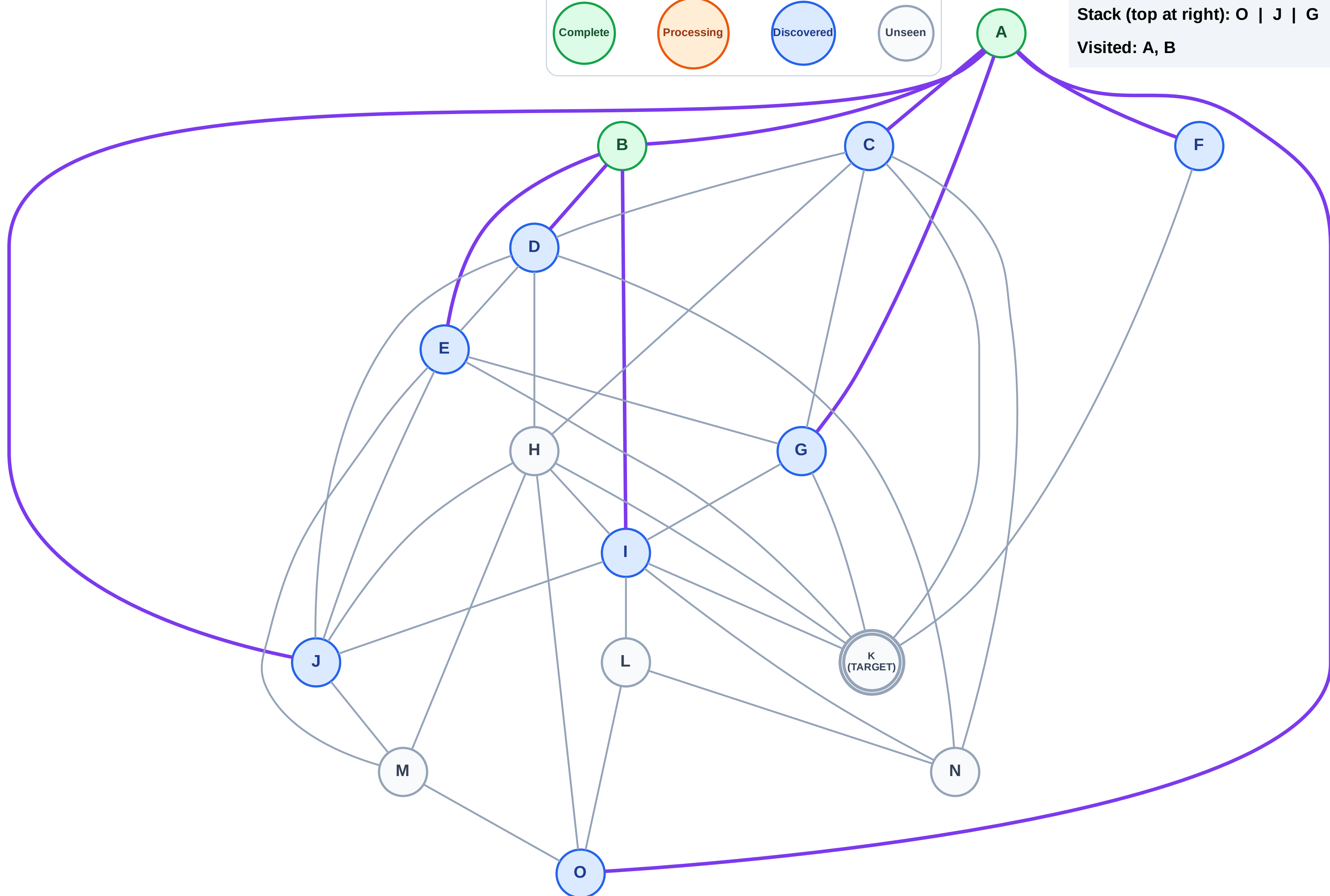
Discovered



Unseen

Stack (top at right): O | J | G | F | C | I | E | D

Visited: A, B



DFS Traversal

Step 6: Process node D

Legend

Complete

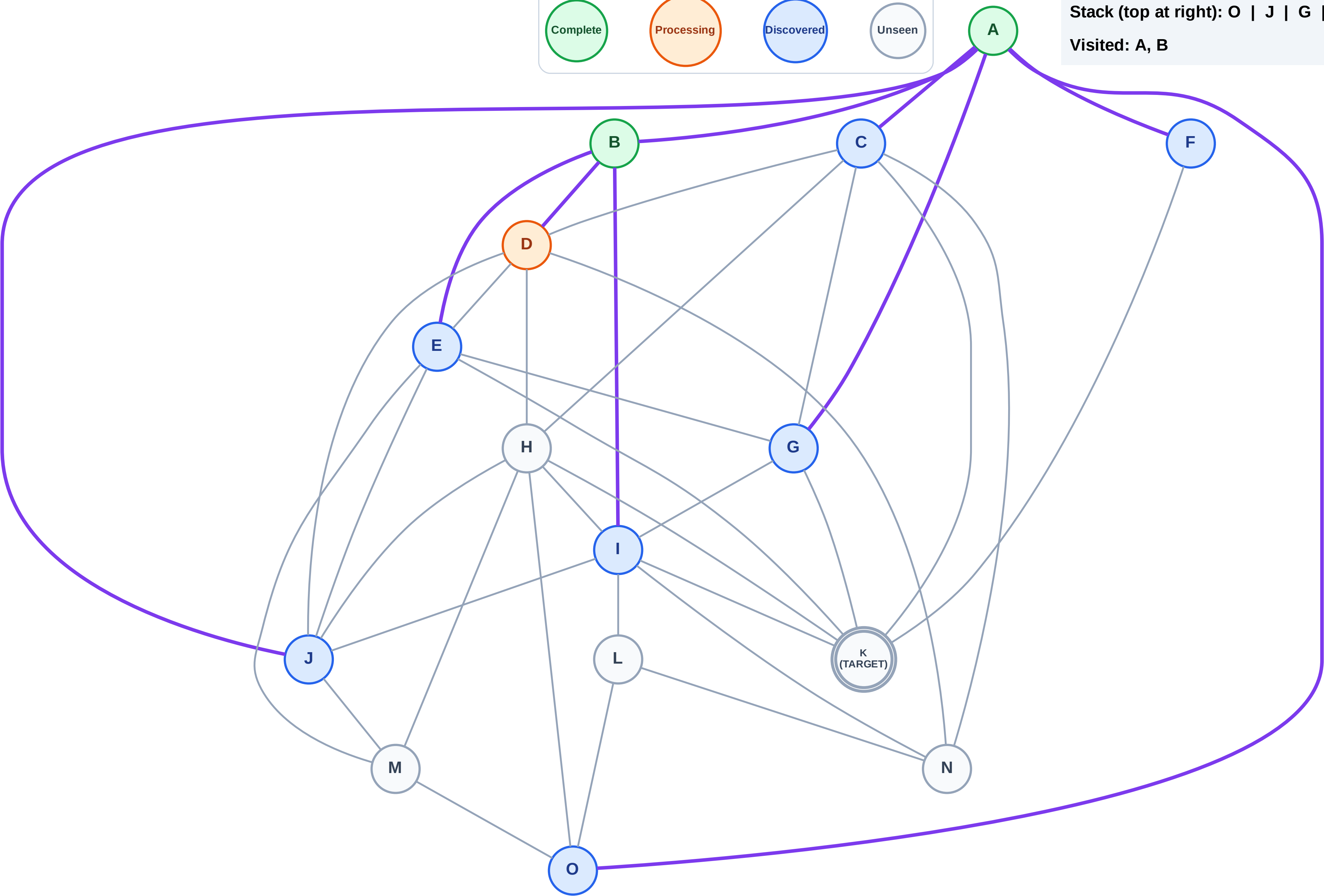
Processing

Discovered

Unseen

Stack (top at right): O | J | G | F | C | I | E

Visited: A, B



Step 7: Discover N, H from D

Step 7: Discover N, H from D

Legend



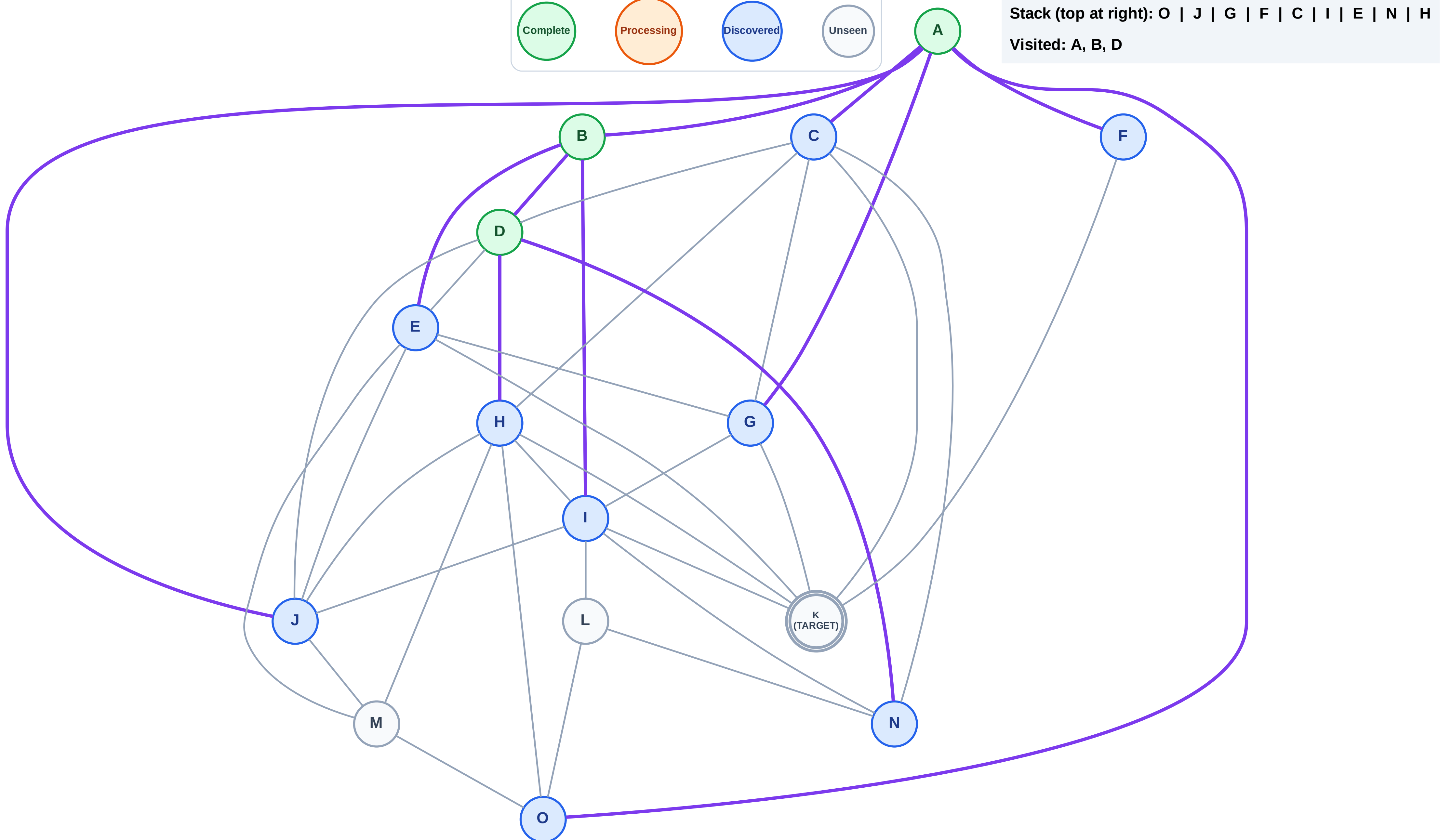
The legend consists of four colored circles arranged horizontally, each containing a label. From left to right: a green circle labeled 'Complete', an orange circle labeled 'Processing', a blue circle labeled 'Discovered', and a grey circle labeled 'Unseen'.

Processing

Unseen

Stack (top at right): O | J | G | F | C | I | E | N | H

Visited: A, B, D



DFS Traversal

Step 8: Process node H

Legend

Complete

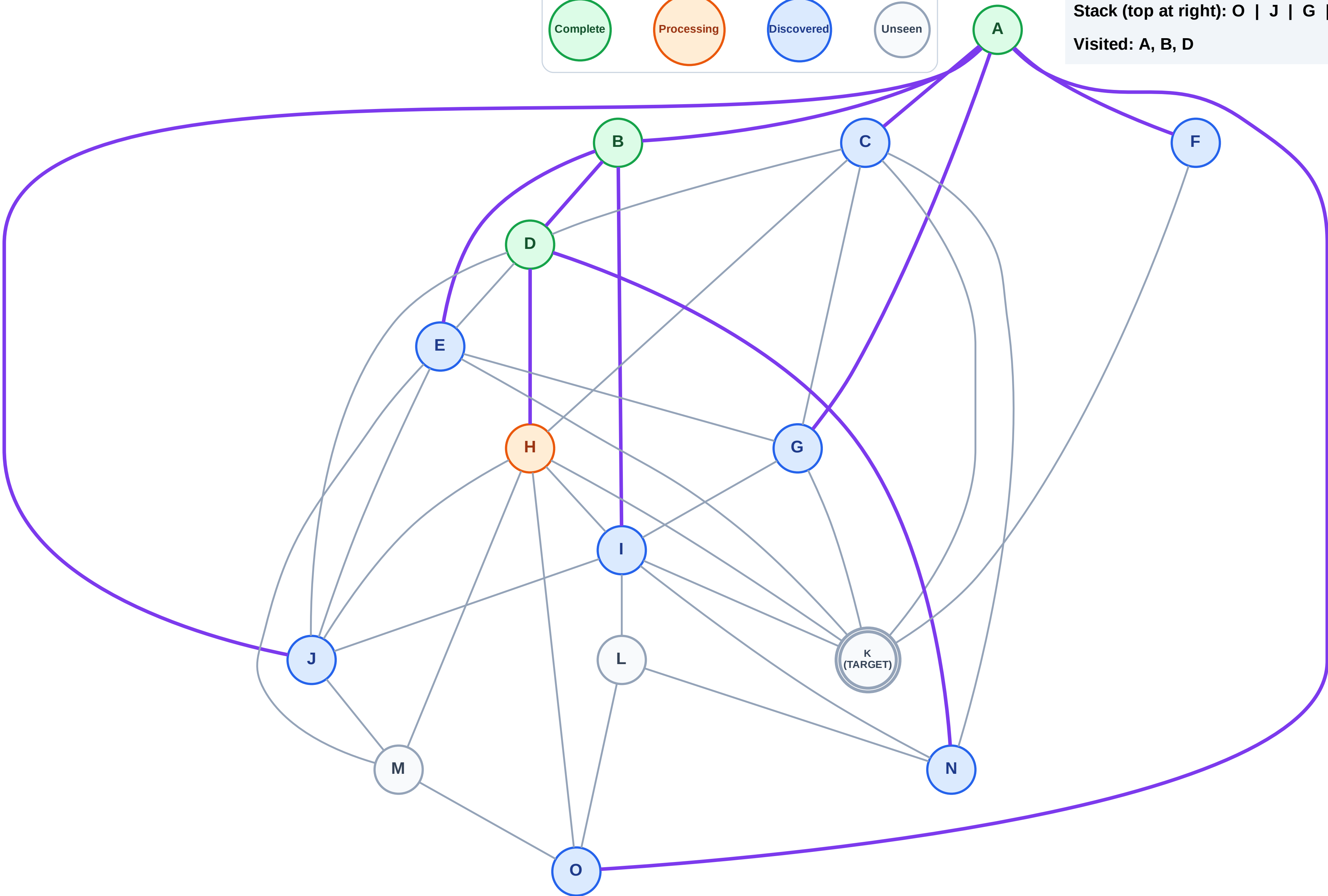
Processing

Discovered

Unseen

Stack (top at right): O | J | G | F | C | I | E | N

Visited: A, B, D



DFS Traversal

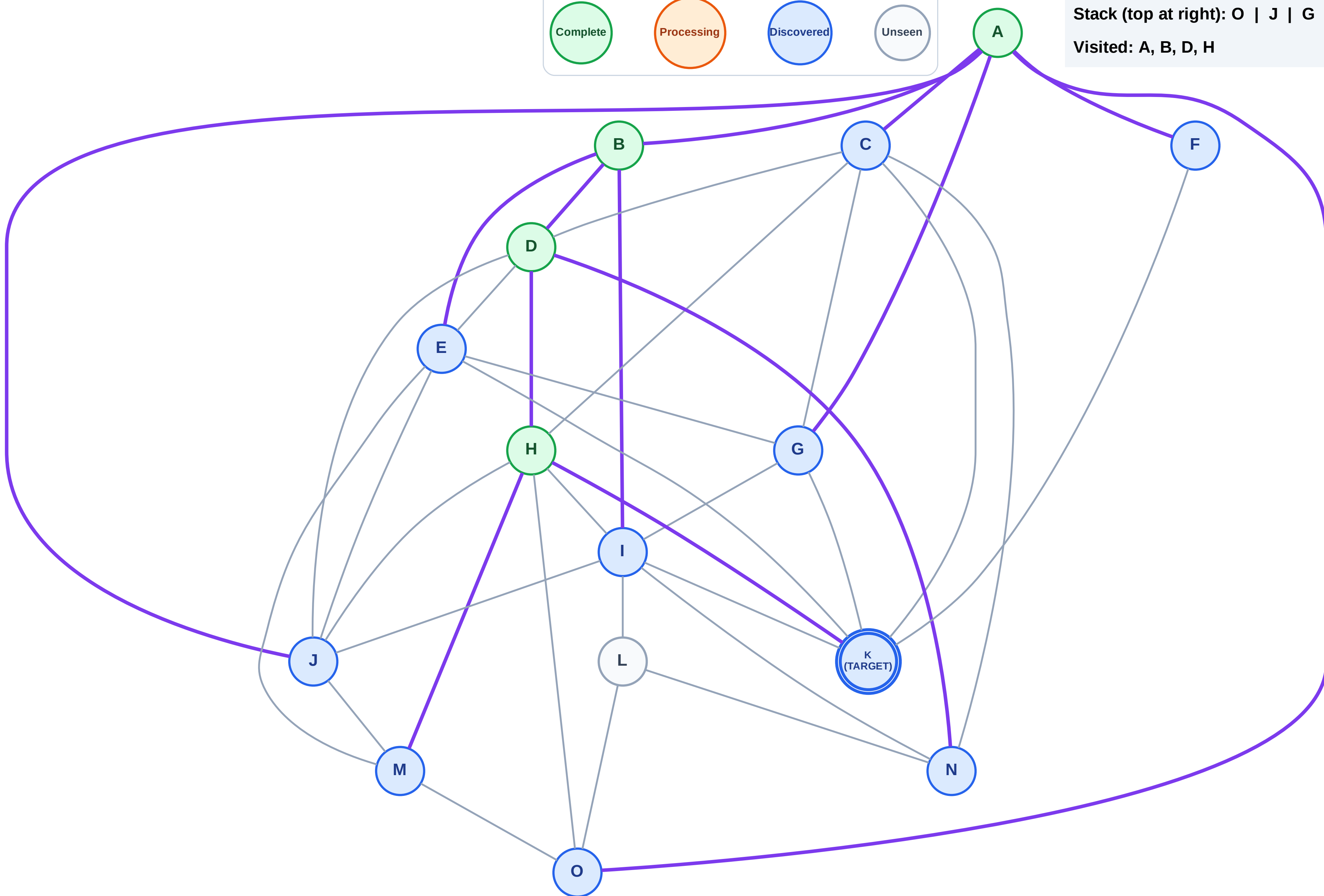
Step 9: Discover M, K from H

Legend



Stack (top at right): O | J | G | F | C | I | E | N | M | K

Visited: A, B, D, H



DFS Traversal

Step 10: Process node K

Legend

Complete

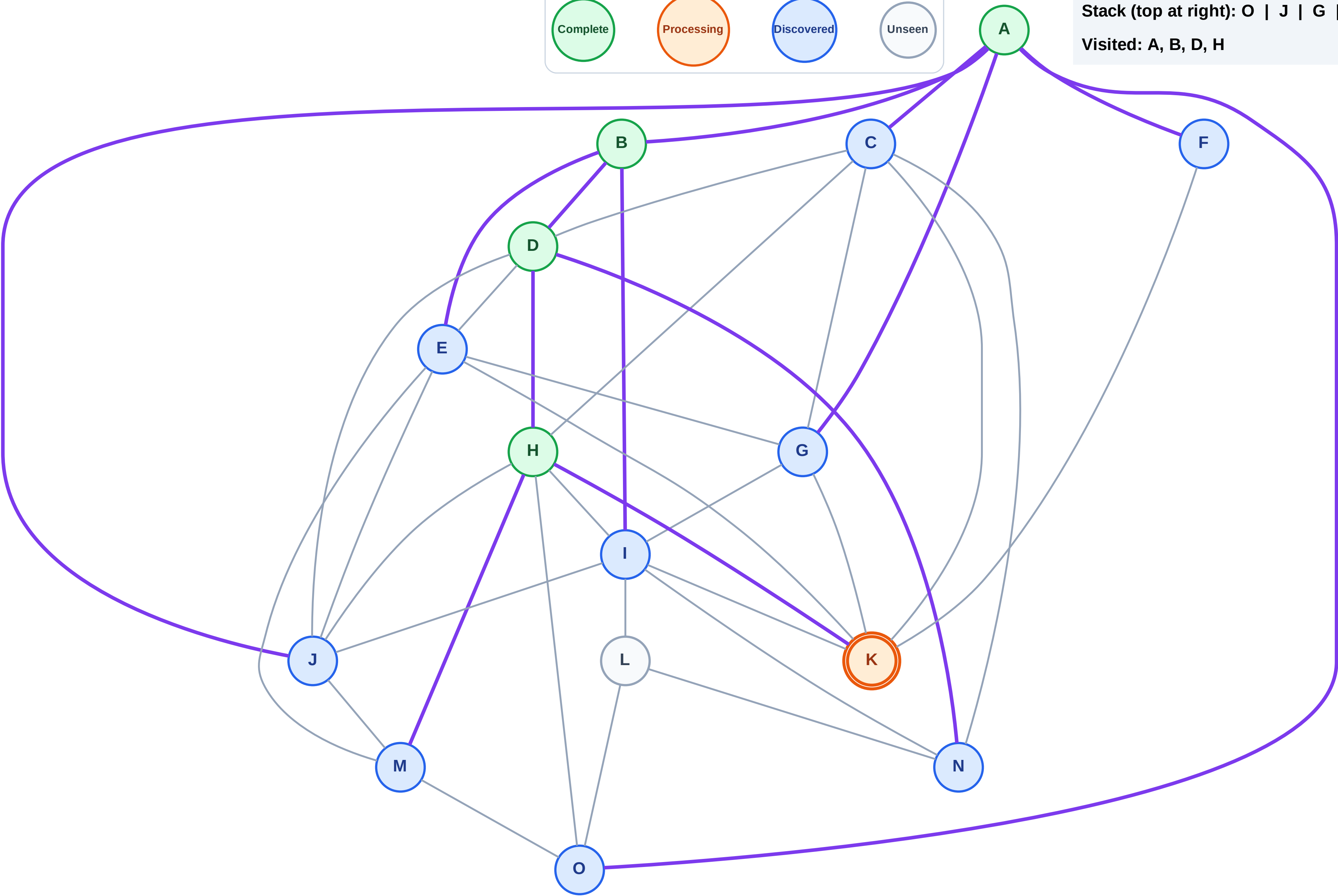
Processing

Discovered

Unseen

Stack (top at right): O | J | G | F | C | I | E | N | M

Visited: A, B, D, H



DFS Traversal

Step 11: TARGET NODE K REACHED! Path (4 hops): A → B → D → H → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): O | J | G | F | C | I | E | N | M

Visited: A, B, D, H, K

